# Inequalities with one unknown

Solving an inequality is solving an equation with one extra rule — and an answer that is a whole interval, not a single number.

LESSON 49–50 | 2 × 40 MIN | ALGEBRA 8, §15 | STAGE 9 · 4.3

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

| | |
|---|---|
| Warm-up | 5 min |
| Explanation | 12 min |
| Interactive | 8 min |
| Practice | 13 min |
| Homework | 2 min |

LEARNING OBJECTIVES

- Solve a linear inequality in one unknown.
- Show the solution on a number line and write it as an interval.
- Reverse the sign when multiplying or dividing by a negative number.
- Recognise an inequality with no solutions or with every number as a solution.

TEXTBOOK

| | |
|---|---|
| National | §15, pp. 85–93 |
| Cambridge | Learner's Book pp. 96–102 |
| Workbook | Workbook 4.3 |

## 01 Explanation

### What a solution is

DEFINITION

A **solution** of an inequality with one unknown is any value of the unknown that makes the inequality a true statement. To **solve** the inequality is to find *all* of them.

An equation such as $2x = 6$ has one answer. The inequality $2x > 6$ has infinitely many — every number bigger than 3. So the answer is written as an **interval**, not a number.

### The method

Exactly the steps you use for an equation, with one addition:

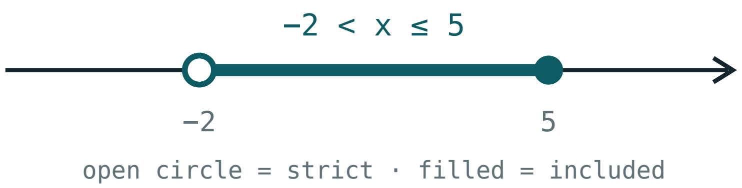
1. Clear fractions and open brackets.
2. Collect the unknowns on one side, the numbers on the other.

3. Divide by the coefficient of $x$ — and if that coefficient is **negative**, **reverse the sign**.
4. Draw the answer on a number line and write the interval.

> $3x - 7 < x + 5 \Longrightarrow 2x < 12 \Longrightarrow x < 6$

> THE ONLY REAL TRAP
>
> From $-4x \geq 12$ you get $x \leq -3$. Write the reversed sign **first**, then do the division — that way you cannot forget it.

## Drawing and writing the answer




*An open circle for a strict inequality, a filled circle when the endpoint is included.*

| INEQUALITY | ON THE LINE | INTERVAL |
|---|---|---|
| $x > 3$ | open circle at 3, shaded right | $(3; +\infty)$ |
| $x \geq 3$ | filled circle at 3, shaded right | $[3; +\infty)$ |
| $x < 3$ | open circle at 3, shaded left | $(-\infty; 3)$ |
| $x \leq 3$ | filled circle at 3, shaded left | $(-\infty; 3]$ |

An infinity end is always written with a round bracket — you never reach it.

## Two special answers

Sometimes the unknown disappears. Then read what is left:

$$2x + 5 > 2x + 1 \Longrightarrow 5 > 1$$

That is true whatever $x$ is, so **every number** is a solution: the answer is $(-\infty; +\infty)$.

$$3x + 2 < 3x - 4 \Longrightarrow 2 < -4$$

That is false whatever $x$ is, so the inequality has **no solutions**.

## 02 Worked examples

EXAMPLE 1 ***Solve $5x - 3 \le 2x + 9$***

1. $5x - 2x \le 9 + 3$

   Collect x on the left, numbers on the right.

2. $3x \le 12$

3. $x \le 4$

   Dividing by 3, a positive number — the sign stays.

4. $(-\infty; 4]$

   Filled circle at 4, shaded to the left.

ANSWER

$x \le 4$, that is $(-\infty; 4]$

EXAMPLE 2 **Solve $4 - 3x > 19$**

1. $-3x > 15$

   Subtract 4 from both sides.

2. Dividing by $-3$ — the sign must reverse.

   Write the new sign first.

3. $x < -5$

4. $(-\infty; -5)$

   Open circle at −5.

ANSWER

$x < -5$

EXAMPLE 3 **Solve $\frac{x - 1}{2} + \frac{x}{3} \geq 1$**

1. *multiply by 6*

   The LCD of 2 and 3 — 6 is positive, so the sign stays.

2. $3(x - 1) + 2x \geq 6$

3. $3x - 3 + 2x \geq 6$

4. $5x \geq 9$, so $x \geq 1.8$

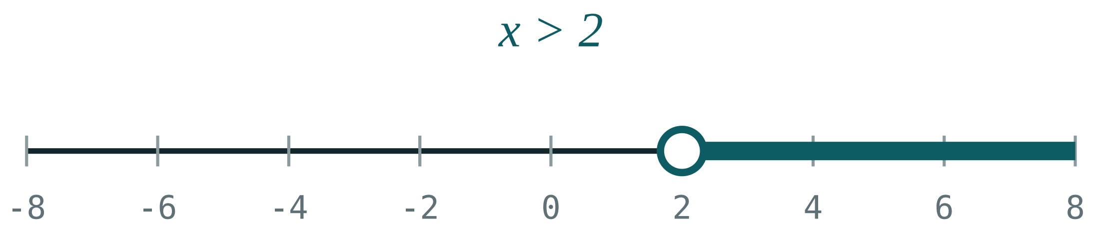
ANSWER

$x \geq 1.8$, that is $[1.8; +\infty)$

## 03 Interactive model

Move the boundary, flip the sign, and toggle strict — three controls cover every answer in this lesson.

### Solutions on the number line




inequality **x > 2** | boundary **2 (open)** | x = 0 works? **no** | x = 6 works? **yes**

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| Inequality with one unknown | Bir noma’lumli tengsizlik | Неравенство с одним неизвестным |
| Solution of an inequality | Tengsizlikning yechimi | Решение неравенства |
| Solution set | Yechimlar to‘plami | Множество решений |
| Interval | Oraliq | Промежуток |
| Number line | Sonlar o‘qi | Числовая прямая |
| Equivalent inequalities | Teng kuchli tengsizliklar | Равносильные неравенства |
| Reverse the sign | Ishorani almashtirish | Изменить знак неравенства |
| No solutions | Yechimga ega emas | Нет решений |
| Every number | Har qanday son | Любое число |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

# 05 Quick check

### Check yourself

$-2x > 10$ gives:

- $x > -5$
- $x < -5$
- $x > 5$
- $x < 5$

$x \geq 3$ as an interval is:

- $(3; +\infty)$
- $[3; +\infty)$
- $(-\infty; 3]$
- $[3; +\infty]$

$2x + 5 > 2x + 1$ has:

- no solutions
- one solution
- every number as a solution
- $x > 0$

$3x + 2 < 3x - 4$ has:

- no solutions
- one solution
- every number
- $x < 0$

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up 7 PROBLEMS

1. *Solve* $2x > 8$

   ANSWER $x > 4$

2. *Solve* $x + 3 \leq 7$

   ANSWER $x \leq 4$

3. *Solve* $x - 5 > 1$

   ANSWER $x > 6$

4. *Solve* $-x < 2$

   ANSWER $x > -2$

5. *Solve* $3x \geq -9$

   ANSWER $x \geq -3$

6. *Solve* $-2x \geq 6$

   ANSWER $x \leq -3$

7. *Write* $x < 5$ *as an interval*

   ANSWER $(-\infty; 5)$

## Medium · the standard question

7 PROBLEMS

1. Solve $5x - 3 \leq 2x + 9$

   ANSWER $x \leq 4$

2. Solve $4 - 3x > 19$

   ANSWER $x < -5$

3. Solve $2(x - 1) < x + 4$

   ANSWER $x < 6$

4. Solve $\frac{x}{3} + 2 \geq 5$

   ANSWER $x \geq 9$

5. Solve $7 - x \leq 2x + 1$

   ANSWER $x \geq 2$

6. Write $x \geq -4$ as an interval

   ANSWER $[-4; +\infty)$

7. Solve $-5x + 2 < 17$

   ANSWER $x > -3$

## Hard · stretch and reason

7 PROBLEMS

1. *Solve* $\dfrac{x-1}{2} + \dfrac{x}{3} \geq 1$

   ANSWER $x \geq 1.8$

2. *Solve* $3(2x - 1) - 2(x + 4) < 5$

   ANSWER $6x - 3 - 2x - 8 < 5$, so $x < 4$

3. *Solve* $2x + 5 > 2(x + 1)$

   ANSWER Every number — $5 > 2$ is always true.

4. *Solve* $3(x - 2) < 3x - 4$

   ANSWER $3x - 6 < 3x - 4$ reduces to $-6 < -4$, which is always true — every number is a solution.

5. *Find the smallest whole number satisfying* $4x - 7 > 9$

   ANSWER $x > 4$, so $x = 5$

6. *Find the largest whole number satisfying* $5 - 2x \geq -3$

   ANSWER $x \leq 4$, so $x = 4$

7. *For which* $x$ *is* $\dfrac{2x-1}{3} - \dfrac{x}{2} \leq 0$?

   ANSWER Multiply by 6: $4x - 2 - 3x \leq 0$, so $x \leq 2$.

## 07 Homework

### Homework — 6 problems

Algebra 8, §15, pp. 85–93. Draw the number line for every answer.

1. *Solve* $4x - 5 \geq x + 7$
2. *Solve* $6 - 5x > 21$
3. *Solve* $3(x + 2) \leq 2x + 10$

4. *Solve* $\frac{x}{4} - 1 < 2$

5. *Find the smallest whole number with* $7x - 3 > 25$

6. *Solve* $2(x - 3) > 2x - 5$ *and explain your answer*

# Systems of inequalities. Number intervals

Two conditions at once — the answer is where the two solution sets overlap.

LESSON 51–52 | 2 × 40 MIN | ALGEBRA 8, §16 | STAGE 9 · 4.3

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

| | |
|---|---|
| Warm-up | 5 min |
| Explanation | 12 min |
| Interactive | 8 min |
| Practice | 13 min |
| Homework | 2 min |

LEARNING OBJECTIVES

- Solve a system of two linear inequalities.
- Find the intersection of two intervals on the number line.
- Use the standard names and notation for intervals.
- Solve a double inequality such as $-2 < 3x + 1 \le 10$.

TEXTBOOK

| | |
|---|---|
| National | §16, pp. 94–104 |
| Cambridge | Learner's Book pp. 96–102 |
| Workbook | Workbook 4.3 |

## 01 Explanation

### The names of the intervals

| CONDITION | NOTATION | NAME |
|---|---|---|
| $a \le x \le b$ | $[a; b]$ | segment (closed interval) |
| $a < x < b$ | $(a; b)$ | open interval |
| $a \le x < b$ | $[a; b)$ | half-open interval |
| $x \ge a$ | $[a; +\infty)$ | closed ray |
| $x < a$ | $(-\infty; a)$ | open ray |

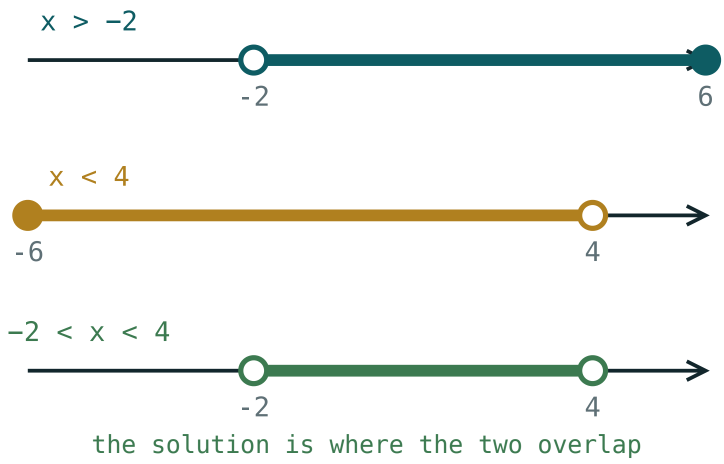
Square bracket = the end belongs to the set (filled circle). Round bracket = it does not (open circle). Infinity is always round.

### Solving a system

METHOD

1. Solve each inequality on its own.

2. Draw both solution sets on the **same** number line.
3. The answer is the part shaded **twice** — the intersection.




*Each inequality shades part of the line; the system is satisfied only where both shadings overlap.*

The word to say in class is **"and"**. A system asks for the values satisfying the first inequality *and* the second, so only the overlap counts.

SOMETIMES THERE IS NO OVERLAP

The system $x > 5$ and $x < 2$ has **no solutions** — the empty set. Say so; do not write a nonsense interval like $(5; 2)$.

## Double inequalities

$-2 < 3x + 1 \le 10$ is a system written compactly. Work on all three parts at once:

$$-2 < 3x + 1 \le 10$$

$$-3 < 3x \le 9 \quad \text{(subtract 1 throughout)}$$

$$-1 < x \le 3 \quad \text{(divide by 3 throughout)}$$

The answer is the half-open interval $(-1; 3]$.

IF YOU DIVIDE BY A NEGATIVE

**Both** signs reverse and the two ends swap places, so rewrite the smaller number on the left.

## 02 Worked examples

**EXAMPLE 1** ***Solve the system:*** $2x - 1 > 3$ ***and*** $x + 4 < 9$

1. $2x > 4$, so $x > 2$

   The first inequality.

2. $x < 5$

   The second.

3. Draw both: shaded right of 2, and left of 5.

4. $2 < x < 5$

   The overlap.

**ANSWER**

$(2; 5)$

**EXAMPLE 2** ***Solve*** $-2 < 3x + 1 \leq 10$

1. $-3 < 3x \leq 9$

   Subtract 1 from all three parts.

2. $-1 < x \leq 3$

   Divide all three by 3.

3. $(-1; 3]$

   Open at −1, closed at 3.

**ANSWER**

$(-1; 3]$

EXAMPLE 3 ***Solve the system: $x \geq 4$ and $2x < 6$***

1. $x \geq 4$

   The first.

2. $x < 3$

   The second.

3. Nothing is both at least 4 and less than 3.

4. No solutions — the empty set.

ANSWER

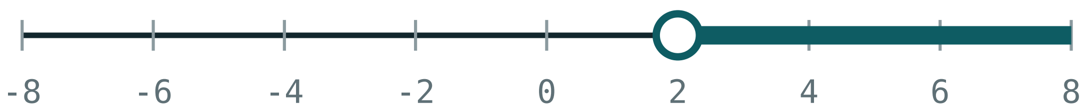
No solutions.

## 03 Interactive model

Set each boundary in turn and ask the class which part of the line survives both.

### One inequality at a time

$$x > 2$$




inequality `x > 2` | boundary `2 (open)` | x = 0 works? **no** | x = 6 works? **yes**

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| System of inequalities | Tengsizliklar sistemasi | Система неравенств |
| Intersection | Kesishma | Пересечение |
| Number interval | Sonli oraliq | Числовой промежуток |
| Segment (closed interval) | Kesma | Отрезок |
| Open interval | Ochiq oraliq | Интервал |
| Half-open interval | Yarim ochiq oraliq | Полуинтервал |
| Ray (half-line) | Nur | Луч |
| Double inequality | Qo‘sh tengsizlik | Двойное неравенство |
| Empty set | Bo‘sh to‘plam | Пустое множество |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

# 05 Quick check

### Check yourself

$[2; 7)$ means:

- $2 < x < 7$
- $2 \le x < 7$
- $2 < x \le 7$
- $2 \le x \le 7$

The system $x > 1$ and $x < 6$ gives:

- $(1; 6)$
- $(-\infty; 6)$
- $(1; +\infty)$
- no solutions

The system $x > 5$ and $x < 2$ gives:

- $(2; 5)$
- $(5; 2)$
- no solutions
- every number

$-4 \le 2x < 6$ gives:

- $[-2; 3)$
- $(-2; 3]$
- $[-4; 6)$
- $[-2; 3]$

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up 7 PROBLEMS

1. *Write $3 \le x \le 8$ as an interval*

   ANSWER $[3; 8]$

2. *Write $0 < x < 5$ as an interval*

   ANSWER $(0; 5)$

3. *Write $x \ge 2$ as an interval*

   ANSWER $[2; +\infty)$

4. *What does $(-1; 4]$ mean?*

   ANSWER $-1 < x \le 4$

5. *Solve the system $x > 1$ and $x < 6$*

   ANSWER $(1; 6)$

6. *Solve the system $x \ge 0$ and $x \le 3$*

   ANSWER $[0; 3]$

7. *Solve the system $x > 4$ and $x > 7$*

   ANSWER $(7; +\infty)$

## Medium · the standard question

7 PROBLEMS

1. *Solve:* $2x - 1 > 3$ *and* $x + 4 < 9$

   ANSWER $(2; 5)$

2. *Solve:* $3x \le 12$ *and* $x + 1 > 0$

   ANSWER $(-1; 4]$

3. *Solve* $-2 < 3x + 1 \le 10$

   ANSWER $(-1; 3]$

4. *Solve* $-4 \le 2x < 6$

   ANSWER $[-2; 3)$

5. *Solve:* $x \ge 4$ *and* $2x < 6$

   ANSWER No solutions.

6. *Solve* $1 \le x - 3 \le 5$

   ANSWER $[4; 8]$

7. *Solve:* $5 - x > 1$ *and* $x + 2 \ge 0$

   ANSWER $[-2; 4)$

## Hard · stretch and reason

7 PROBLEMS

1. *Solve* $-5 < 2 - 3x \le 8$

   ANSWER $-7 < -3x \le 6 \Longrightarrow -2 \le x < \frac{7}{3}$

2. *Solve:* $2(x - 1) \ge x$ *and* $3x < x + 8$

   ANSWER $x \ge 2$ and $x < 4$: $[2; 4)$

3. *How many whole numbers satisfy* $-3 < 2x + 1 \le 9$?

   ANSWER $-2 < x \le 4$: the numbers $-1, 0, 1, 2, 3, 4$ — six of them.

4. *Solve:* $\frac{x}{2} > 1$ *and* $\frac{x}{3} < 2$

   ANSWER $x > 2$ and $x < 6$: $(2; 6)$

5. *For which a does the system* $x > 3$ *and* $x < a$ *have solutions?*

   ANSWER $a > 3$

6. *Solve* $0 \le \frac{2x - 1}{3} \le 3$

   ANSWER $0 \le 2x - 1 \le 9 \Longrightarrow [0.5; 5]$

7. *Find all whole numbers satisfying both* $4x > 6$ *and* $3x \le 15$

   ANSWER $x > 1.5$ and $x \le 5$: $2, 3, 4, 5$

## 07 Homework

### Homework — 6 problems

Algebra 8, §16, pp. 94–104. Draw both sets on one line before writing the answer.

1. *Solve:* $3x - 2 > 7$ *and* $x - 1 < 6$
2. *Solve* $-6 \le 4x + 2 < 10$
3. *Solve:* $x + 5 \ge 2$ *and* $2x \le 8$
4. *Solve:* $x > 6$ *and* $x < 3$, *and explain the answer*

5. *How many whole numbers satisfy $-4 < 3x - 1 \le 11$?*

6. *Write $[-3; 5)$ in words and draw it on a number line*

# The modulus of a number

Distance from zero — one idea that turns every modulus equation and inequality into something you can already solve.

LESSON 53–54 | 2 × 40 MIN | ALGEBRA 8, §17 | STAGE 9 · BEYOND

LESSON CLOCK


| | |
|---|---|
| Warm-up | 5 min |
| Explanation | 12 min |
| Interactive | 8 min |
| Practice | 13 min |
| Homework | 2 min |

LEARNING OBJECTIVES

- Define $|\,a\,|$ and evaluate it.
- Read $|\,x\,|$ as the distance from $x$ to 0 on the number line.
- Solve equations of the form $|\,x - a\,| = b$.
- Solve inequalities $|\,x\,| < b$ and $|\,x\,| > b$.

TEXTBOOK

| | |
|---|---|
| National | §17, pp. 105–110 |
| Cambridge | Extension beyond Stage 9 |
| Workbook | – |

## 01 Explanation

### The definition

DEFINITION

$$|\,a\,| = a \text{ if } a \geq 0, \quad |\,a\,| = -a \text{ if } a < 0$$

So $|\,5\,| = 5$, $|\,-5\,| = 5$, $|\,0\,| = 0$. The modulus is **never negative**.

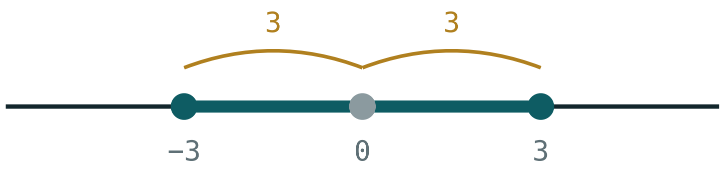
−A IS NOT A NEGATIVE NUMBER

If $a = -5$ then $-a = 5$. The second line of the definition *removes* the minus sign; it does not add one.

Two facts worth stating at once: $|\,a\,| \geq 0$ always, and $|\,a\,| = |\,-a\,|$ — opposite numbers have the same modulus.

### The picture: distance from zero

$|\,a\,|$ is the **distance** from the point $a$ to the origin. That single sentence solves the whole topic.




$|x| \le 3$ means $-3 \le x \le 3$

*Every point within 3 of the origin satisfies $|x| \le 3$ — the segment from −3 to 3.*

More generally, $|x - a|$ is the distance from $x$ to $a$. So $|x - 4| = 3$ asks: which points are exactly 3 away from 4? Answer: $1$ and $7$.

## Equations

RULE

For $b > 0$: $|A| = b$ means $A = b$ **or** $A = -b$.
$|A| = 0$ means $A = 0$. $|A| = b$ with $b < 0$ has **no solutions**.

$$|x - 4| = 3 \Longrightarrow x - 4 = 3 \text{ or } x - 4 = -3 \Longrightarrow x = 7 \text{ or } x = 1$$

Always check the sign of the right-hand side first. $|x + 2| = -5$ needs no working at all — a modulus cannot be negative.

## Inequalities

TWO SHAPES, TWO ANSWERS

$$|x| < b \text{ means } -b < x < b \text{ — one interval, in the middle}$$

$$|x| > b \text{ means } x < -b \textbf{ or } x > b \text{ — two rays, outside}$$

The picture makes it obvious: “less than $b$ from zero” is the middle piece; “more than $b$ from zero” is everything outside.

Same with a shift: $|x - 5| \le 2$ means the distance from 5 is at most 2, so $3 \le x \le 7$.

## 02 Worked examples

### EXAMPLE 1 Solve $|2x - 1| = 7$

1. $2x - 1 = 7$ or $2x - 1 = -7$

   Two cases.

2. $2x = 8$, so $x = 4$

   First case.

3. $2x = -6$, so $x = -3$

   Second case.

4. check: $|7| = 7$ ✓ and $|-7| = 7$ ✓

ANSWER

$x = 4$ or $x = -3$

### EXAMPLE 2 Solve $|x - 3| < 5$

1. The distance from 3 is less than 5.

   Read it as distance.

2. $-5 < x - 3 < 5$

   The middle-piece form.

3. $-2 < x < 8$

   Add 3 throughout.

ANSWER

$(-2; 8)$

EXAMPLE 3 ***Solve* $|\,x + 1\,| \geq 4$**

1. The distance from $-1$ is at least 4.
   This is the “outside” shape.

2. $x + 1 \leq -4$ or $x + 1 \geq 4$
   Two rays.

3. $x \leq -5$ or $x \geq 3$

ANSWER

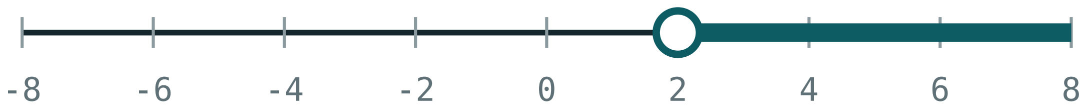
$(-\infty; -5] \cup [3; +\infty)$

## 03 Interactive model

Set the boundary at 3 and switch between < and > — the two answer shapes appear side by side.

**Inside or outside?**

$x > 2$




inequality **`x > 2`** | boundary **`2 (open)`** | x = 0 works? **no** | x = 6 works? **yes**

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| Modulus (absolute value) | Modul (absolyut qiymat) | Модуль (абсолютная величина) |
| Distance from zero | Noldan uzoqlik | Расстояние от нуля |
| Non-negative | Manfiy bo‘lmagan | Неотрицательный |
| Two cases | Ikki hol | Два случая |
| Opposite numbers | Qarama-qarshi sonlar | Противоположные числа |
| Equation with a modulus | Modulli tenglama | Уравнение с модулем |
| Inequality with a modulus | Modulli tengsizlik | Неравенство с модулем |
| Solution set | Yechimlar to‘plami | Множество решений |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

# 05 Quick check

### Check yourself

$|-7|$ equals:

- $-7$
- $7$
- $0$
- $\pm 7$

$|x| = 5$ gives:

- $x = 5$
- $x = -5$
- $x = 5 \text{ or } x = -5$
- no solutions

$|x| < 4$ means:

- $x < 4$
- $-4 < x < 4$
- $x > 4$
- $x < -4 \text{ or } x > 4$

$|x + 2| = -3$ has:

- one solution
- two solutions
- no solutions
- every number

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up

7 PROBLEMS

1. *Find* $|\ 9\ |$

   ANSWER $9$

2. *Find* $|\ -9\ |$

   ANSWER $9$

3. *Find* $|\ 0\ |$

   ANSWER $0$

4. *Find* $|\ -3\ | + |\ 4\ |$

   ANSWER $7$

5. *Solve* $|\ x\ | = 6$

   ANSWER $x = 6$ or $x = -6$

6. *Solve* $|\ x\ | = 0$

   ANSWER $x = 0$

7. *Solve* $|\ x\ | < 2$

   ANSWER $-2 < x < 2$

## Medium · the standard question

7 PROBLEMS

1. *Solve* $|x - 4| = 3$

ANSWER $x = 7$ or $x = 1$

2. *Solve* $|2x - 1| = 7$

ANSWER $x = 4$ or $x = -3$

3. *Solve* $|x + 2| = 5$

ANSWER $x = 3$ or $x = -7$

4. *Solve* $|x - 3| < 5$

ANSWER $(-2; 8)$

5. *Solve* $|x| \geq 3$

ANSWER $x \leq -3$ or $x \geq 3$

6. *Solve* $|x + 1| \geq 4$

ANSWER $x \leq -5$ or $x \geq 3$

7. *Solve* $|x + 2| = -5$

ANSWER No solutions.

## Hard · stretch and reason

7 PROBLEMS

1. *Solve* $|3x + 6| = 12$

   ANSWER $x = 2$ or $x = -6$

2. *Solve* $|x - 5| \le 2$

   ANSWER $[3; 7]$

3. *Solve* $|2x - 3| < 5$

   ANSWER $-1 < x < 4$

4. *Solve* $|1 - x| = 4$

   ANSWER $x = -3$ or $x = 5$

5. *How many whole numbers satisfy* $|x| \le 4$*?*

   ANSWER Nine: $-4$ to $4$.

6. *Solve* $|x - 2| > 0$

   ANSWER Every $x$ except $x = 2$.

7. *Find the distance between the points* $x = -3$ *and* $x = 8$ *using a modulus*

   ANSWER $|8 - (-3)| = 11$

## 07 Homework

### Homework — 6 problems

Algebra 8, §17, pp. 105–110. Draw the number line for the inequalities.

1. *Find* $|-12|$, $|12|$ *and* $|-12| - |5|$
2. *Solve* $|x - 6| = 4$
3. *Solve* $|3x + 1| = 10$
4. *Solve* $|x| \le 7$
5. *Solve* $|x - 1| > 3$

6. *Explain why* $| x + 3 | = -2$ *has no solutions*

Home / Grade 8 / Algebra / Quarter 3

# Approximate values and error

Every measurement is approximate — this lesson says by how much, and how to write it honestly.

LESSON 55–57 | 3 × 40 MIN | ALGEBRA 8, §§18–19 | STAGE 9 · 3.4

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

| | |
|---|---|
| Warm-up | 5 min |
| Explanation | 12 min |
| Interactive | 8 min |
| Practice | 13 min |
| Homework | 2 min |

LEARNING OBJECTIVES

- Distinguish an exact value from an approximate one.
- Calculate the absolute error and the relative error.
- Round to a given number of decimal places or significant figures.
- Say which of two measurements is the more accurate.

TEXTBOOK

| | |
|---|---|
| National | §18 pp. 111–113, §19 pp. 114–117 |
| Cambridge | Learner’s Book pp. 74–81 |
| Workbook | Workbook 3.4 |

## 01 Explanation

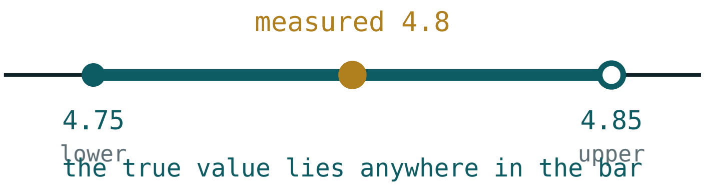
### Exact and approximate

A count is exact: there are exactly 27 learners in the class. A measurement is not: a table is “1.2 m long” only to the accuracy of the tape. Every measured number is an **approximate value** of some exact value we shall never know.

DEFINITION

If $a$ is the exact value and $x$ an approximation of it, the **absolute error** is $\Delta = |\, a - x \,|$.

We rarely know $\Delta$ exactly — we know a **bound** for it. Writing $x = 4.8 \pm 0.05$ says the true value lies between $4.75$ and $4.85$.




*A measurement of 4.8 rounded to one decimal place: the true value lies anywhere in the bar.*

## Relative error

An error of 1 cm matters enormously on a pencil and not at all on a road. What decides is the error **compared with the size** of the thing measured.

DEFINITION

$$\textit{relative error} = \frac{\Delta}{|x|}, \text{ usually written as a percentage}$$

1 cm on a 20 cm pencil is $\frac{1}{20} = 5\%$; 1 cm on a 1000 cm road is $0.1\%$. The second measurement is a hundred times more accurate, even though the absolute error is the same.

TO COMPARE TWO MEASUREMENTS

Compare the **relative** errors. The smaller relative error is the more accurate measurement.

## Rounding

**To a decimal place.** Look at the first digit you are dropping: 5 or more, round up; less than 5, leave it. $3.147 \to 3.15$ to 2 d.p.

**To significant figures.** Count from the first non-zero digit. $0.004736 \to 0.0047$ to 2 s.f.; $52\,800 \to 53\,000$ to 2 s.f.

THE ERROR A ROUNDING CREATES

Rounding to the nearest $0.1$ gives an absolute error of at most $0.05$ — half the last unit. To the nearest whole number, at most $0.5$.

ROUND ONCE, AT THE END

Rounding part-way through a calculation and then again at the end makes the error grow. Keep the full figures until the last line.

## 02 Worked examples

EXAMPLE 1 ***The exact value is $a = 3.14159$ and the approximation is $x = 3.14$. Find both errors.***

1. $\Delta = |\,3.14159 - 3.14\,| = 0.00159$

   Absolute error.

2. $\dfrac{0.00159}{3.14} \approx 0.00051$

   Divide by the approximation.

3. $\approx 0.05\%$

   As a percentage.

ANSWER

$\Delta \approx 0.0016$, relative error $\approx 0.05\%$

EXAMPLE 2

**A length is measured as 4.8 m to one decimal place. Between which values does the true length lie?**

1. One decimal place means the nearest $0.1$.
2. The error is at most half of that: $0.05$.
3. $4.8 - 0.05 = 4.75$ and $4.8 + 0.05 = 4.85$
4. $4.75 \le length < 4.85$

   The lower bound is included, the upper is not.

ANSWER

$4.75\ m \le \ell < 4.85\ m$

EXAMPLE 3

**Which is more accurate: 12 cm measured to the nearest cm, or 250 cm measured to the nearest cm?**

1. Both have the same absolute error, at most $0.5\ cm$.
2. $\frac{0.5}{12} \approx 4.2\%$

   First relative error.
3. $\frac{0.5}{250} = 0.2\%$

   Second relative error.
4. The second is far more accurate.

   The same error on a bigger quantity matters less.

ANSWER

The 250 cm measurement.

## 03 Interactive model

Ask for a bound before any calculation — “the answer is somewhere between …”.

### Errors and rounding

Exact $a = 3.14159$, approximation $x = 3.14$.

1. $\Delta = | a - x | = 0.00159$

Absolute error — just the difference, without a sign.

2. $\dfrac{\Delta}{x} = \dfrac{0.00159}{3.14} \approx 0.00051$

Relative error.

3. $\approx 0.05\%$

Multiply by 100 to state it as a percentage.

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| Exact value | Aniq qiymat | Точное значение |
| Approximate value | Taqribiy qiymat | Приближённое значение |
| Absolute error | Absolyut xatolik | Абсолютная погрешность |
| Relative error | Nisbiy xatolik | Относительная погрешность |
| Round | Yaxlitlash | Округление |
| Decimal place | O‘nli xona | Десятичный знак |
| Significant figures | Ahamiyatli raqamlar | Значащие цифры |
| Accuracy | Aniqlik | Точность |
| Measurement | O‘lchash | Измерение |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

# 05 Quick check

### Check yourself

The absolute error of an approximation is:

- $a - x$
- $|a - x|$
- $\frac{a}{x}$
- $a + x$

A length rounded to 4.8 m (1 d.p.) lies between:

- $4.7$ and $4.9$
- $4.75$ and $4.85$
- $4.79$ and $4.81$
- $4.8$ exactly

$0.004736$ to 2 s.f. is:

- $0.00$
- $0.0047$
- $0.0048$
- $0.004$

To compare the accuracy of two measurements you use:

- the absolute error
- the relative error
- the larger value
- the number of digits

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up 7 PROBLEMS

1. *Round 3.847 to 1 d.p.*

   ANSWER *3.8*

2. *Round 3.847 to 2 d.p.*

   ANSWER *3.85*

3. *Round 5629 to 2 s.f.*

   ANSWER *5600*

4. *Round 0.00382 to 2 s.f.*

   ANSWER *0.0038*

5. *Exact 7, approximation 6.9. Find the absolute error.*

   ANSWER *0.1*

6. *Exact 50, approximation 49. Find the absolute error.*

   ANSWER *1*

7. *A length is 6 m to the nearest metre. Find the largest possible error.*

   ANSWER *0.5 m*

## Medium · the standard question

7 PROBLEMS

1. *Exact 3.14159, approximation 3.14. Find the absolute error.*

   ANSWER $\approx 0.0016$

2. *The same numbers: find the relative error as a percentage.*

   ANSWER $\approx 0.05\%$

3. *A length reads 4.8 m to 1 d.p. Give the bounds.*

   ANSWER $4.75 \le \ell < 4.85$

4. *A mass reads 250 g to the nearest 10 g. Give the bounds.*

   ANSWER $245 \le m < 255$

5. *Exact 20, approximation 19.5. Find the relative error.*

   ANSWER $2.5\%$

6. *Which is more accurate: 12 cm or 250 cm, both to the nearest cm?*

   ANSWER $250$ cm — relative error $0.2\%$ against $4.2\%$.

7. *Round 0.0004736 to 3 s.f.*

   ANSWER $0.000474$

## Hard · stretch and reason

7 PROBLEMS

1. *A rectangle is 4.2 m by 3.5 m, both to 1 d.p. Find the bounds of its perimeter.*

   ANSWER $4.15 \le a < 4.25$, $3.45 \le b < 3.55$, so $15.2 \le P < 15.6$

2. *The same rectangle: find the bounds of its area.*

   ANSWER $14.3175 \le S < 15.0875$

3. *A measurement has absolute error at most 0.02 and relative error 0.5%. Estimate the value.*

   ANSWER $\frac{0.02}{x} = 0.005$, so $x = 4$

4. *Two masses: 3.5 kg ± 0.05 and 480 g ± 5 g. Which is more accurate?*

   ANSWER *1.43%* against *1.04%* — the 480 g measurement.

5. *Explain why rounding twice can be worse than rounding once.*

   ANSWER Each rounding adds its own error, and the errors accumulate: $2.46 \to 2.5 \to 3$ but $2.46 \to 2$ directly.

6. *A number rounds to 7.30 to 2 d.p. Give its bounds.*

   ANSWER $7.295 \le x < 7.305$

7. *A speed of 60 km/h is measured to the nearest 5 km/h. Find the relative error at most.*

   ANSWER $\frac{2.5}{60} \approx 4.2\%$

## 07 Homework

### Homework — 6 problems

Algebra 8, §§18–19, pp. 111–117.

1. *Round 2.7182 to 1, 2 and 3 decimal places.*
2. *Round 34 782 to 2 and to 3 significant figures.*
3. *Exact 8, approximation 7.6. Find the absolute and relative errors.*
4. *A length reads 12.3 cm to 1 d.p. Give the bounds.*

5. *Which is more accurate: 5 m to the nearest metre, or 500 m to the nearest metre?*

6. *A square has side 6.4 cm to 1 d.p. Find the bounds of its perimeter.*

# Upper and lower bounds

The national revision and practical lessons, carrying the Cambridge Stage 9 treatment of bounds in calculations.

LESSON 58–59 | 2 × 40 MIN | ALGEBRA 8, CHAPTER II REVISION | STAGE 9 · 3.4 [CAMBRIDGE INSERT]

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

| | |
|---|---|
| Warm-up | 5 min |
| Explanation | 12 min |
| Interactive | 8 min |
| Practice | 13 min |
| Homework | 2 min |

LEARNING OBJECTIVES

- Write the upper and lower bound of a rounded measurement.
- Find the bounds of a sum and of a difference.
- Find the bounds of a product and of a quotient.
- Decide how many figures an answer may honestly be given to.

TEXTBOOK

| | |
|---|---|
| National | pp. 118–121 |
| Cambridge | Learner's Book pp. 76–81 |
| Workbook | Workbook 3.4 |

## 01 Explanation

### The bounds of a rounded number

RULE

A value rounded to the nearest $u$ lies within $\frac{u}{2}$ of the stated figure:

$$x - \frac{u}{2} \le \textit{true value} < x + \frac{u}{2}$$

| STATED | ROUNDED TO | LOWER BOUND | UPPER BOUND |
|---|---|---|---|
| $7\ cm$ | nearest cm | $6.5$ | $7.5$ |
| $4.8\ m$ | 1 d.p. | $4.75$ | $4.85$ |
| $250\ g$ | nearest 10 g | $245$ | $255$ |
| $3.60\ s$ | 2 d.p. | $3.595$ | $3.605$ |

The lower bound is reached; the upper bound is not. That is why it is written with ≤ on the left and $<$ on the right.

## Bounds of a sum, difference, product, quotient

WHICH BOUND WITH WHICH

| | LARGEST ANSWER | SMALLEST ANSWER |
|---|---|---|
| $a + b$ | $UB(a) + UB(b)$ | $LB(a) + LB(b)$ |
| $a - b$ | $UB(a) - LB(b)$ | $LB(a) - UB(b)$ |
| $a \times b$ | $UB(a) \times UB(b)$ | $LB(a) \times LB(b)$ |
| $a \div b$ | $UB(a) \div LB(b)$ | $LB(a) \div UB(b)$ |

(for positive quantities)

The pattern is worth saying aloud: for a **sum** or a **product**, both go the same way; for a **difference** or a **quotient**, they go opposite ways — to make a difference as large as possible, take the most you can and subtract the least you can.

## How many figures may the answer have?

Work out both bounds, then keep only the digits on which they agree.

If the bounds of an area come out as $14.3175$ and $15.0875$, the two do not even agree on the units digit — so the honest answer is “about 15”, not $14.7\ m^2$. Reporting more figures than the bounds support is the commonest fault in practical work.

## 02 Worked examples

**EXAMPLE 1** **A rectangle measures 4.2 m by 3.5 m, each to 1 d.p. Find the bounds of the perimeter.**

1. $4.15 \le a < 4.25$ and $3.45 \le b < 3.55$

   Half of 0.1 either way.

2. $P = 2(a + b)$

3. $LB = 2(4.15 + 3.45) = 15.2$

   Both lower bounds.

4. $UB = 2(4.25 + 3.55) = 15.6$

   Both upper bounds.

ANSWER

$15.2\ m \le P < 15.6\ m$

**EXAMPLE 2** **The same rectangle: find the bounds of the area.**

1. $LB = 4.15 \times 3.45 = 14.3175$

   Both lower bounds.

2. $UB = 4.25 \times 3.55 = 15.0875$

   Both upper bounds.

3. The bounds disagree already in the units digit.

4. So the area is “about 15 m²” — one significant figure is all that is justified.

ANSWER

$14.32\ m^2 \le S < 15.09\ m^2$, so $\approx 15\ m^2$

EXAMPLE 3

**A car travels 120 km (to the nearest km) in 2.0 hours (to 1 d.p.). Find the greatest possible average speed.**

1. $119.5 \le d < 120.5$

   Distance bounds.

2. $1.95 \le t < 2.05$

   Time bounds.

3. For the greatest speed: most distance, least time.

   A quotient — opposite bounds.

4. $v < \dfrac{120.5}{1.95} \approx 61.8\ km/h$

ANSWER

$\approx 61.8\ km/h$

## 03 Interactive model

Ask for the two extreme cases before any arithmetic — the rule then writes itself.

**Which bound goes with which?**

$a = 7 \pm 0.5$, $b = 4 \pm 0.5$. Largest $a + b$?

1. A sum grows when either part grows.

So both go up.

2. $UB(a) = 7.5$

, $UB(b) = 4.5$

3. $a + b < 12$

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| Upper bound | Yuqori chegara | Верхняя граница |
| Lower bound | Quyi chegara | Нижняя граница |
| Rounded to | Yaxlitlangan | Округлено до |
| Nearest | Eng yaqin | Ближайший |
| Half a unit | Birlikning yarmi | Половина единицы |
| Sum | Yig‘indi | Сумма |
| Difference | Ayirma | Разность |
| Product | Ko‘paytma | Произведение |
| Quotient | Bo‘linma | Частное |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

# 05 Quick check

### Check yourself

$7\ cm$ to the nearest cm has bounds:

- $6.9$ and $7.1$
- $6.5$ and $7.5$
- $6$ and $8$
- $6.95$ and $7.05$

For the largest value of $a - b$ you take:

- both upper bounds
- both lower bounds
- UB(a) and LB(b)
- LB(a) and UB(b)

For the largest value of $a \div b$ you take:

- UB(a) ÷ UB(b)
- UB(a) ÷ LB(b)
- LB(a) ÷ LB(b)
- LB(a) ÷ UB(b)

$250\ g$ to the nearest 10 g has lower bound:

- $249.5$
- $245$
- $240$
- $249$

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up

7 PROBLEMS

1. *9 cm to the nearest cm. Give the bounds.*

   ANSWER $8.5 \le \ell < 9.5$

2. *3.2 m to 1 d.p. Give the bounds.*

   ANSWER $3.15 \le \ell < 3.25$

3. *80 kg to the nearest 10 kg. Give the bounds.*

   ANSWER $75 \le m < 85$

4. *2.50 s to 2 d.p. Give the bounds.*

   ANSWER $2.495 \le t < 2.505$

5. $a = 5 \pm 0.5$, $b = 3 \pm 0.5$. *Largest* $a + b$.

   ANSWER $9$

6. *Same values: smallest* $a + b$.

   ANSWER $7$

7. *600 m to the nearest 100 m. Give the bounds.*

   ANSWER $550 \le d < 650$

## Medium · the standard question

7 PROBLEMS

1. $a = 7 \pm 0.5$, $b = 4 \pm 0.5$. Largest $a - b$.

   ANSWER $4$

2. Same values: smallest $a - b$.

   ANSWER $2$

3. A rectangle 4.2 m by 3.5 m (1 d.p.). Bounds of the perimeter.

   ANSWER $15.2 \le P < 15.6$

4. The same rectangle: bounds of the area.

   ANSWER $14.3175 \le S < 15.0875$

5. $d = 120$ km $\pm 0.5$, $t = 2.0$ h $\pm 0.05$. Largest speed.

   ANSWER $\approx 61.8$ km/h

6. The same values: smallest speed.

   ANSWER $\approx 58.3$ km/h

7. A square has side 8 cm to the nearest cm. Bounds of the area.

   ANSWER $56.25 \le S < 72.25$

## Hard · stretch and reason

7 PROBLEMS

1. *A cube has edge 5 cm to the nearest cm. Bounds of its volume.*

   ANSWER $91.125 \le V < 166.375$

2. *To how many significant figures may that volume be given?*

   ANSWER None safely — the bounds do not even share the first digit; report "between 91 and 167 cm³".

3. $a = 12.4 \pm 0.05$, $b = 3.1 \pm 0.05$. *Bounds of* $\frac{a}{b}$

   ANSWER $\frac{12.35}{3.15} \approx 3.92$ to $\frac{12.45}{3.05} \approx 4.08$

4. *A triangle has base 9 cm and height 6 cm, both to the nearest cm. Bounds of its area.*

   ANSWER $\frac{1}{2} \cdot 8.5 \cdot 5.5 = 23.375$ to $\frac{1}{2} \cdot 9.5 \cdot 6.5 = 30.875$

5. *A runner covers 400 m (nearest m) in 52 s (nearest s). Bounds of the average speed.*

   ANSWER $\frac{399.5}{52.5} \approx 7.61$ to $\frac{400.5}{51.5} \approx 7.78$ m/s

6. *Explain why the upper bound uses* $<$ *and the lower bound uses* $\le$.

   ANSWER A value exactly on the lower bound rounds up to the stated figure; one exactly on the upper bound rounds up to the next figure instead.

7. *Two lengths are 5.0 cm and 5.00 cm. Which is stated more precisely, and why?*

   ANSWER *5.00 cm* — it claims the nearest $0.01$, so bounds $4.995$ to $5.005$, ten times tighter.

## 07 Homework

### Homework — 6 problems

Algebra 8, pp. 118–121, and Cambridge Learner's Book 3.4.

1. *15 cm to the nearest cm. Give the bounds.*
2. *6.70 kg to 2 d.p. Give the bounds.*
3. $a = 20 \pm 0.5$, $b = 8 \pm 0.5$. *Find the bounds of* $a - b$.

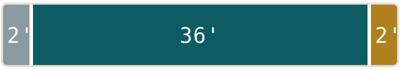
4. *A rectangle is 7.5 cm by 4.2 cm, both to 1 d.p. Find the bounds of the area.*
5. *A car covers 250 km (nearest km) in 4.0 h (1 d.p.). Find the bounds of the average speed.*
6. *A square has side 12 cm to the nearest cm. Find the bounds of the perimeter.*

# Control work 5 · Inequalities, modulus and accuracy

The Chapter II assessment, then a work-on-mistakes lesson sorted by error type.

LESSON 60–61 | 2 × 40 MIN | ALGEBRA 8, §§15–19 | STAGE 9 · 3.4, 4.3

**LESSON CLOCK**




| | |
|---|---|
| Setting up | 2 min |
| The paper | 36 min |
| Collect in | 2 min |

**LEARNING OBJECTIVES**

- Assess inequalities in one unknown, systems, modulus, error and bounds.
- Sort every lost mark into one of four error types.
- Re-solve every task that was lost.

**TEXTBOOK**

| | |
|---|---|
| National | Revision of §§15–19 |
| Cambridge | Learner's Book pp. 74–81, 96–102 |
| Workbook | Workbook 3.4, 4.3 |

## 01 Explanation

### Lesson 60 — the paper (40 minutes)

Two variants of seven tasks, 2 marks each:

- tasks 1–2 · solve a linear inequality, one with a negative coefficient (§15)
- task 3 · solve a system and write the interval (§16)
- task 4 · a double inequality (§16)
- task 5 · a modulus equation (§17)
- task 6 · a modulus inequality (§17)
- task 7 · bounds or relative error (§§18–19)

### Lesson 61 — work on mistakes (40 minutes)

**THE FOUR ERRORS THIS PAPER PRODUCES**

1. **The sign not reversed** after dividing by a negative number.
2. **Only one case taken** for a modulus equation — the negative case forgotten.
3. **The two modulus inequality shapes confused** — writing two rays where one interval belongs, or the reverse.

4. **Bracket types mixed up** when writing an interval, or ∞ given a square bracket.

## 02 Worked examples

EXAMPLE 1 **Find the error: $|x - 2| = 5$, so $x = 7$**

1. Only the positive case was taken.
   This is error 2.

2. $x - 2 = 5$ or $x - 2 = -5$
   Both cases.

3. $x = 7$ or $x = -3$

ANSWER

$x = 7$ or $x = -3$

EXAMPLE 2 **Find the error: $|x| > 3$, so $-3 < x < 3$**

1. That is the answer for $|x| < 3$, not for $>$.
   The two shapes are swapped.

2. “More than 3 from zero” is everything **outside**.

3. $x < -3$ or $x > 3$

ANSWER

$(-\infty; -3) \cup (3; +\infty)$

## 03 Interactive model

Show the wrong working, take a vote on the error type, then reveal.

### Diagnose the error

**Claimed:** $-3x < 12 \Longrightarrow x < -4$

1. Dividing by $-3$ — a negative number.

2. The sign turns: $x > -4$.

3. Test $x = 0$: $0 < 12$ ✓, and $0 > -4$ ✓.

One test settles it.

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| Control work | Nazorat ishi | Контрольная работа |
| Work on mistakes | Xatolar ustida ishlash | Работа над ошибками |
| Solution set | Yechimlar to‘plami | Множество решений |
| Interval | Oraliq | Промежуток |
| Modulus | Modul | Модуль |
| Bound | Chegara | Граница |
| Error | Xato | Ошибка |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

# 05 Quick check

### Check yourself

$-3x < 12$ gives:

- $x < -4$
- $x > -4$
- $x < 4$
- $x > 4$

$| x - 2 | = 5$ gives:

- $x = 7$
- $x = -3$
- $x = 7$ or $x = -3$
- no solutions

$| x | > 3$ gives:

- $-3 < x < 3$
- $x < -3$ or $x > 3$
- $x > 3$
- $x < 3$

$x \geq 2$ as an interval is:

- $(2;\ +\infty)$
- $[2;\ +\infty)$
- $(2;\ +\infty]$
- $[2;\ +\infty]$

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up 7 PROBLEMS

1. ***Variant 1, task 1.*** *Solve* $4x - 5 > 7$

   ANSWER $x > 3$

2. ***Task 2.*** *Solve* $-2x + 1 \geq 9$

   ANSWER $x \leq -4$

3. ***Task 3.*** *Solve:* $x > 2$ *and* $x \leq 6$

   ANSWER $(2; 6]$

4. ***Task 4.*** *Solve* $-3 < 2x + 1 \leq 7$

   ANSWER $(-2; 3]$

5. ***Task 5.*** *Solve* $|x - 1| = 4$

   ANSWER $x = 5$ or $x = -3$

6. ***Task 6.*** *Solve* $|x| \leq 5$

   ANSWER $[-5; 5]$

7. ***Task 7.*** *8 m to the nearest metre. Give the bounds.*

   ANSWER $7.5 \leq \ell < 8.5$

## Medium · the standard question

7 PROBLEMS

1. ***Variant 2, task 1.*** *Solve* $5x + 2 \le 17$

   ANSWER $x \le 3$

2. ***Task 2.*** *Solve* $3 - 4x > 15$

   ANSWER $x < -3$

3. ***Task 3.*** *Solve:* $x \ge -1$ *and* $x < 4$

   ANSWER $[-1; 4)$

4. ***Task 4.*** *Solve* $-6 \le 3x < 9$

   ANSWER $[-2; 3)$

5. ***Task 5.*** *Solve* $|\, 2x + 1 \,| = 9$

   ANSWER $x = 4$ or $x = -5$

6. ***Task 6.*** *Solve* $|\, x - 2 \,| > 3$

   ANSWER $x < -1$ or $x > 5$

7. ***Task 7.*** *Exact 40, approximation 39. Find the relative error.*

   ANSWER $2.5\%$

## Hard · stretch and reason

7 PROBLEMS

1. *Find the error:* $-3x < 12$, *so* $x < -4$

   ANSWER The sign was not reversed. Correct: $x > -4$.

2. *Find the error:* $|\,x - 2\,| = 5$, *so* $x = 7$

   ANSWER Only one case. Correct: $x = 7$ or $x = -3$.

3. *Find the error:* $|\,x\,| > 3$, *so* $-3 < x < 3$

   ANSWER The shapes are swapped. Correct: $x < -3$ or $x > 3$.

4. *Find the error:* $x \geq 2$ *written as* $(2;\ +\infty]$

   ANSWER Correct: $[2;\ +\infty)$.

5. *Find the error:* $|\,x + 4\,| = -1$, *so* $x = -5$ *or* $x = -3$

   ANSWER A modulus is never negative — there are no solutions.

6. *Find the error: "7 cm to the nearest cm lies between 6.9 and 7.1"*

   ANSWER The bounds are $6.5$ and $7.5$ — half a unit, not a tenth.

7. *Find the error: "*$x > 5$ *and* $x < 2$ *gives* $(5;\ 2)$*"*

   ANSWER The sets do not overlap; there are no solutions.

## 07 Homework

### After the work-on-mistakes lesson

Re-solve every task you lost marks on. Quadratic equations begin next lesson.

1. *Solve* $-5x + 3 \geq 18$
2. *Solve* $|\,3x - 2\,| = 10$
3. *Solve* $|\,x + 1\,| < 6$
4. *A length reads 9.4 cm to 1 d.p. Give the bounds.*
5. Write out the four errors from this lesson in your own words with an example each.

# The quadratic equation and its roots

The first equation of the course whose graph is not a line — and the first that can have two answers, one, or none at all.

LESSON 62–63 | 2 × 40 MIN | ALGEBRA 8, CHAPTER III §20 | STAGE 9 · BEYOND

LESSON CLOCK


| | |
|---|---|
| Warm-up | 5 min |
| Explanation | 12 min |
| Interactive | 8 min |
| Practice | 13 min |
| Homework | 2 min |

LEARNING OBJECTIVES

- Recognise a quadratic equation and name its coefficients.
- Check whether a given number is a root.
- Solve a quadratic by factorising.
- Read the number of roots off the graph.

TEXTBOOK

| | |
|---|---|
| National | Chapter III, pp. 122–128 |
| Cambridge | Extension beyond Stage 9 |
| Workbook | – |

## 01 Explanation

### What a quadratic equation is

DEFINITION

An equation of the form

$$ax^2 + bx + c = 0, \quad a \neq 0$$

is a **quadratic equation**. Here $a$ is the **leading coefficient**, $b$ the second coefficient and $c$ the **free term**.

The condition $a \neq 0$ matters: without it the $x^2$ disappears and the equation is linear.

| EQUATION | A | B | C |
|---|---|---|---|
| $x^2 - 5x + 6 = 0$ | 1 | −5 | 6 |
| $3x^2 + 2x - 8 = 0$ | 3 | 2 | −8 |
| $2x^2 - 7 = 0$ | 2 | 0 | −7 |
| $x^2 + 4x = 0$ | 1 | 4 | 0 |

Write the equation in this standard order **before** reading off the coefficients — and mind the signs: in $x^2 - 5x + 6$ the coefficient $b$ is $-5$, not $5$.

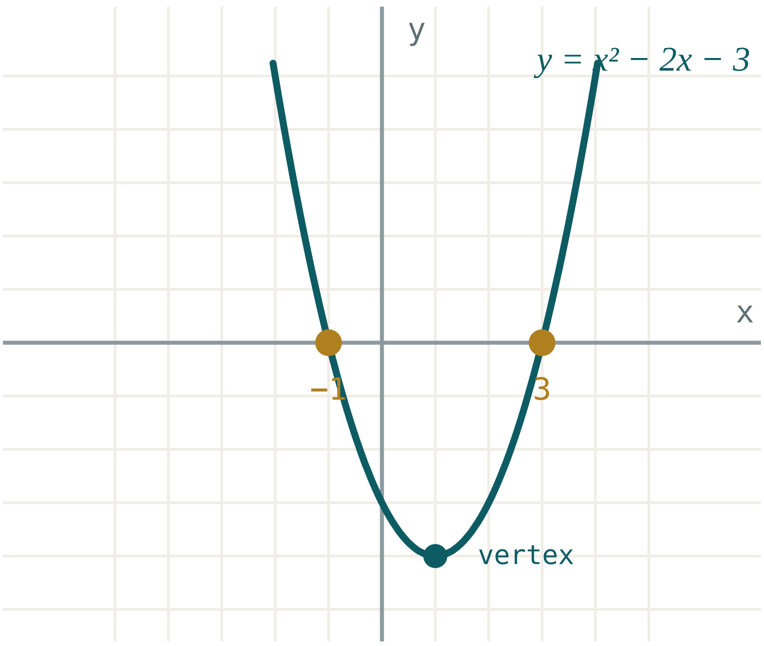
## Roots

DEFINITION

A **root** of the equation is a value of $x$ that makes it true.

To check, substitute. Is $3$ a root of $x^2 - 5x + 6 = 0$? $9 - 15 + 6 = 0$ ✓ — yes.

Graphically, the roots are where the parabola $y = ax^2 + bx + c$ crosses the $x$-axis.




*The graph of y = x² − 2x − 3 crosses the x-axis at −1 and 3 — the two roots of x² − 2x − 3 = 0.*

A parabola can cross the axis twice, touch it once, or miss it entirely — so a quadratic equation has **two roots, one root, or none**. That is the single biggest difference from the linear equations of Grade 7.

# Solving by factorising

THE ZERO PRODUCT RULE

If $A \cdot B = 0$ then $A = 0$ or $B = 0$. Nothing else can happen — this is the only reason factorising solves equations.

So write the quadratic as a product equal to zero, then set each bracket to zero in turn:

$$x^2 - 5x + 6 = 0$$

$$(x - 2)(x - 3) = 0$$

$$x - 2 = 0 \text{ or } x - 3 = 0 \Longrightarrow x = 2 \text{ or } x = 3$$

To factorise $x^2 + px + q$, find two numbers with **product $q$** and **sum $p$**. For $x^2 - 5x + 6$: product 6, sum $-5 \rightarrow -2$ and $-3$.

THE RULE NEEDS A ZERO ON THE RIGHT

From $(x - 1)(x - 4) = 6$ you may **not** write $x - 1 = 6$. Expand, move everything to one side, and factorise again.

# 02 Worked examples

## EXAMPLE 1 *Solve $x^2 - 7x + 12 = 0$*

1. Two numbers with product 12 and sum $-7$.

   Both negative, since the product is positive and the sum negative.

2. $-3$ and $-4$

3. $(x - 3)(x - 4) = 0$

4. $x = 3$ or $x = 4$

   Check: $9 - 21 + 12 = 0$ ✓

ANSWER

$x = 3, x = 4$

## EXAMPLE 2 *Solve $x^2 + 2x - 15 = 0$*

1. Product $-15$, sum $2$.

   Opposite signs, since the product is negative.

2. $5$ and $-3$

3. $(x + 5)(x - 3) = 0$

4. $x = -5$ or $x = 3$

ANSWER

$x = -5, x = 3$

EXAMPLE 3 ***Is $x = -2$ a root of $2x^2 + 3x - 2 = 0$?***

1. $2(-2)^2 + 3(-2) - 2$

   Substitute.

2. $= 8 - 6 - 2$

   Mind the sign: $(-2)^2 = 4$, not $-4$.

3. $= 0$

   It works.

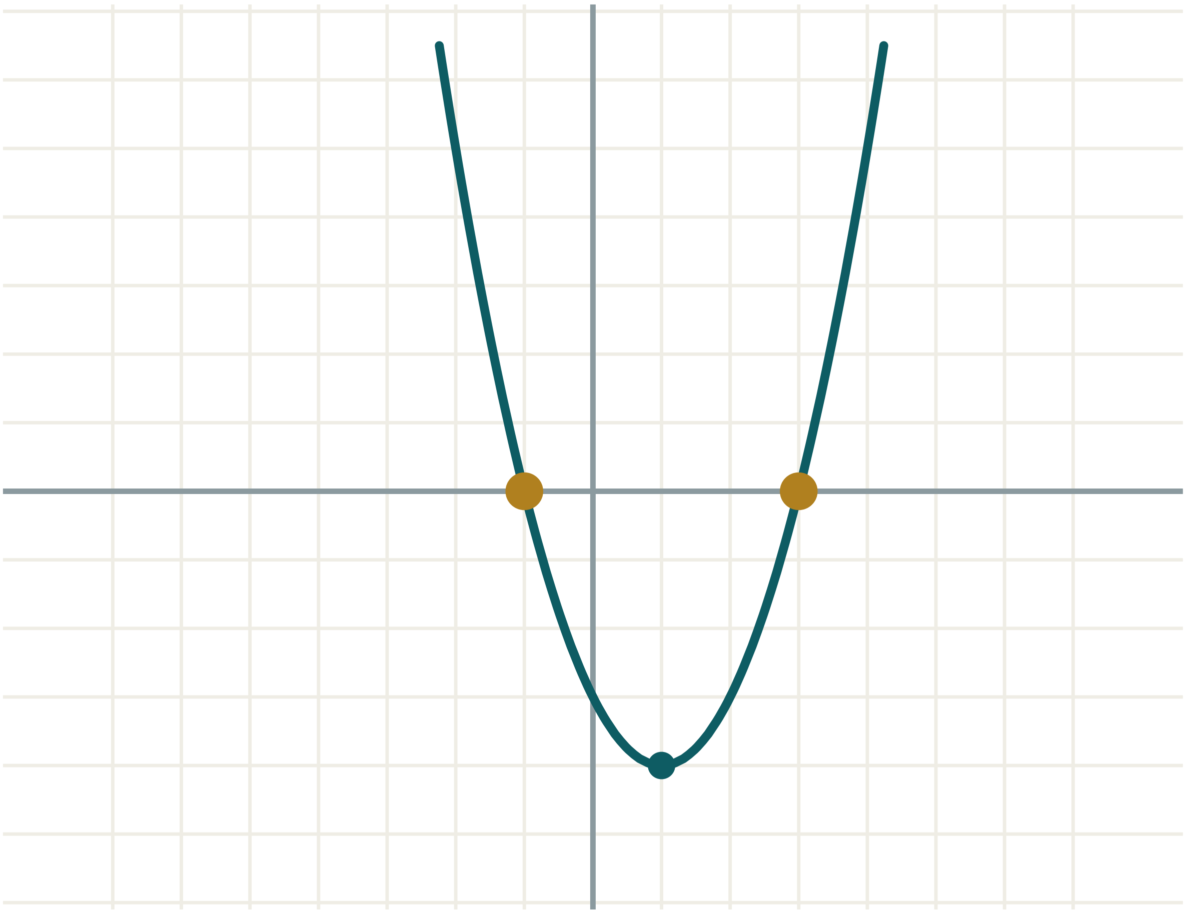
ANSWER

Yes, $x = -2$ is a root.

## 03 Interactive model

Set a = 1, b = −2, c = −3 and read the roots off the graph before doing any algebra.

### Roots and the discriminant



| | | | | | | | |
|---|---|---|---|---|---|---|---|
| a **1** | b **-2** | c **-3** | $D = b^2 - 4ac$ **16** | roots **two** | $x_1$ **-1** | $x_2$ **3** | $x_1 + x_2$ **2** |
| $x_1 \cdot x_2$ **-3** | | | | | | | |

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| Quadratic equation | Kvadrat tenglama | Квадратное уравнение |
| Root of an equation | Tenglamaning ildizi | Корень уравнения |
| Coefficient | Koeffitsiyent | Коэффициент |
| Leading coefficient | Bosh koeffitsiyent | Старший коэффициент |
| Free term (constant) | Ozod had | Свободный член |
| Factorise | Ko‘paytuvchilarga ajratish | Разложить на множители |
| Zero product rule | Ko‘paytmaning nolga tengligi | Произведение равно нулю |
| Parabola | Parabola | Парабола |
| Substitute and check | O‘rniga qo‘yib tekshirish | Подставить и проверить |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

## 05 Quick check

### Check yourself

In $3x^2 - 5x + 2 = 0$, the coefficient b is:

- $3$
- $5$
- $-5$
- $2$

A quadratic equation can have:

- exactly one root
- exactly two roots
- two, one or no roots
- always two roots

$(x - 2)(x + 5) = 0$ gives:

- $x = 2, x = 5$
- $x = -2, x = 5$
- $x = 2, x = -5$
- $x = -2, x = -5$

$x^2 - 5x + 6 = 0$ factorises to:

- $(x - 2)(x - 3)$
- $(x + 2)(x + 3)$
- $(x - 1)(x - 6)$
- $(x - 2)(x + 3)$

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up

7 PROBLEMS

1. *Name a, b, c in* $x^2 + 3x - 4 = 0$

   ANSWER $1, 3, -4$

2. *Name a, b, c in* $2x^2 - 5 = 0$

   ANSWER $2, 0, -5$

3. *Is* $x = 2$ *a root of* $x^2 - 4 = 0$?

   ANSWER Yes.

4. *Is* $x = 1$ *a root of* $x^2 + x - 2 = 0$?

   ANSWER Yes.

5. *Solve* $(x - 3)(x + 1) = 0$

   ANSWER $x = 3, x = -1$

6. *Solve* $(x + 4)(x + 2) = 0$

   ANSWER $x = -4, x = -2$

7. *Solve* $x(x - 5) = 0$

   ANSWER $x = 0, x = 5$

## Medium · the standard question

7 PROBLEMS

1. Solve $x^2 - 7x + 12 = 0$

ANSWER $x = 3, x = 4$

2. Solve $x^2 + 2x - 15 = 0$

ANSWER $x = -5, x = 3$

3. Solve $x^2 - x - 6 = 0$

ANSWER $x = 3, x = -2$

4. Solve $x^2 + 7x + 10 = 0$

ANSWER $x = -2, x = -5$

5. Solve $x^2 - 9x + 20 = 0$

ANSWER $x = 4, x = 5$

6. Is $x = -2$ a root of $2x^2 + 3x - 2 = 0$?

ANSWER Yes.

7. Solve $x^2 - 2x - 8 = 0$

ANSWER $x = 4, x = -2$

## Hard · stretch and reason

7 PROBLEMS

1. *Solve* $2x^2 - 5x + 3 = 0$ *by factorising*

   ANSWER $(2x - 3)(x - 1) = 0$, so $x = 1.5, x = 1$

2. *Solve* $3x^2 + 5x - 2 = 0$

   ANSWER $(3x - 1)(x + 2) = 0$, so $x = \frac{1}{3}, x = -2$

3. *Solve* $(x - 1)(x - 4) = 6$

   ANSWER $x^2 - 5x - 2 = 0$ — does not factorise nicely; the roots come from the formula next lesson.

4. *One root of* $x^2 + bx - 6 = 0$ *is 2. Find b and the other root.*

   ANSWER $4 + 2b - 6 = 0$ gives $b = 1$; the other root is $-3$.

5. *For which c does* $x^2 - 6x + c = 0$ *have* $x = 2$ *as a root?*

   ANSWER $4 - 12 + c = 0$, so $c = 8$

6. *Solve* $x^2 = 5x$

   ANSWER $x(x - 5) = 0$: $x = 0$ or $x = 5$. Never divide by $x$ — you would lose the root 0.

7. *Write a quadratic equation whose roots are 3 and −7.*

   ANSWER $(x - 3)(x + 7) = 0$, i.e. $x^2 + 4x - 21 = 0$

## 07 Homework

### Homework — 6 problems

Algebra 8, Chapter III, pp. 122–128. Check every root by substituting.

1. *Name a, b, c in* $4x^2 - 7x + 1 = 0$
2. *Solve* $x^2 - 8x + 15 = 0$
3. *Solve* $x^2 + 5x - 14 = 0$
4. *Solve* $x^2 = 7x$

5. *Is $x = -3$ a root of $x^2 + x - 6 = 0$?*

6. *Write a quadratic equation with roots 2 and $-5$*

# Incomplete quadratic equations

When b or c is zero the formula is unnecessary — three short methods that are quicker and safer.

LESSON 64–65 | 2 × 40 MIN | ALGEBRA 8, CHAPTER III §21 | STAGE 9 · BEYOND

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

| | |
|---|---|
| Warm-up | 5 min |
| Explanation | 12 min |
| Interactive | 8 min |
| Practice | 13 min |
| Homework | 2 min |

LEARNING OBJECTIVES

- Recognise the three incomplete forms.
- Solve $ax^2 + c = 0$ by taking roots.
- Solve $ax^2 + bx = 0$ by taking out a common factor.
- Explain why you must never divide an equation by $x$.

TEXTBOOK

| | |
|---|---|
| National | Chapter III, pp. 129–133 |
| Cambridge | Extension beyond Stage 9 |
| Workbook | – |

## 01 Explanation

### The three forms

| FORM | MISSING | METHOD | ROOTS |
|---|---|---|---|
| $ax^2 = 0$ | b and c | divide by a | $x = 0$ only |
| $ax^2 + bx = 0$ | c | common factor x | $x = 0$ and $x = -\dfrac{b}{a}$ |
| $ax^2 + c = 0$ | b | make $x^2$ the subject | two, or none |

### Form 1 — no b

$$ax^2 + c = 0 \Longrightarrow x^2 = -\frac{c}{a}$$

- If $-\dfrac{c}{a} > 0$ there are **two roots**, $x = \pm\sqrt{-c/a}$.
- If it is $0$, one root, $x = 0$.
- If it is **negative**, there are **no roots** — no real number squares to a negative.

$$2x^2 - 18 = 0 \Longrightarrow x^2 = 9 \Longrightarrow x = \pm 3$$

$$x^2 + 4 = 0 \Longrightarrow x^2 = -4 \Longrightarrow \text{no roots}$$

DO NOT LOSE THE MINUS ROOT

$x^2 = 9$ has **two** answers, $3$ and $-3$. The symbol $\sqrt{9}$ means only $3$ — that is why the ± has to be written by hand.

## Form 2 — no c

$$ax^2 + bx = 0 \Longrightarrow x(ax + b) = 0 \Longrightarrow x = 0 \text{ or } x = -\frac{b}{a}$$

$$3x^2 - 12x = 0 \Longrightarrow 3x(x - 4) = 0 \Longrightarrow x = 0 \text{ or } x = 4$$

NEVER DIVIDE BY X

Dividing $3x^2 = 12x$ by $x$ gives $3x = 12$, so $x = 4$ — and the root $x = 0$ has vanished. You may only divide by something you know is not zero, and $x$ might be. Always move everything to one side and factorise instead.

An equation of this form **always** has $x = 0$ as a root, because the free term is zero.

## 02 Worked examples

**EXAMPLE 1** ***Solve $5x^2 - 45 = 0$***

1. $5x^2 = 45$

2. $x^2 = 9$

   Divide by 5.

3. $x = 3$ or $x = -3$

   Both square roots.

ANSWER

$x = \pm 3$

**EXAMPLE 2** ***Solve $4x^2 + 7x = 0$***

1. $x(4x + 7) = 0$

   Common factor x.

2. $x = 0$ or $4x + 7 = 0$

   Zero product rule.

3. $x = 0$ or $x = -1.75$

ANSWER

$x = 0, x = -1.75$

EXAMPLE 3 ***Solve $3x^2 + 12 = 0$***

1. $3x^2 = -12$
2. $x^2 = -4$
3. A square is never negative.
4. No real roots.

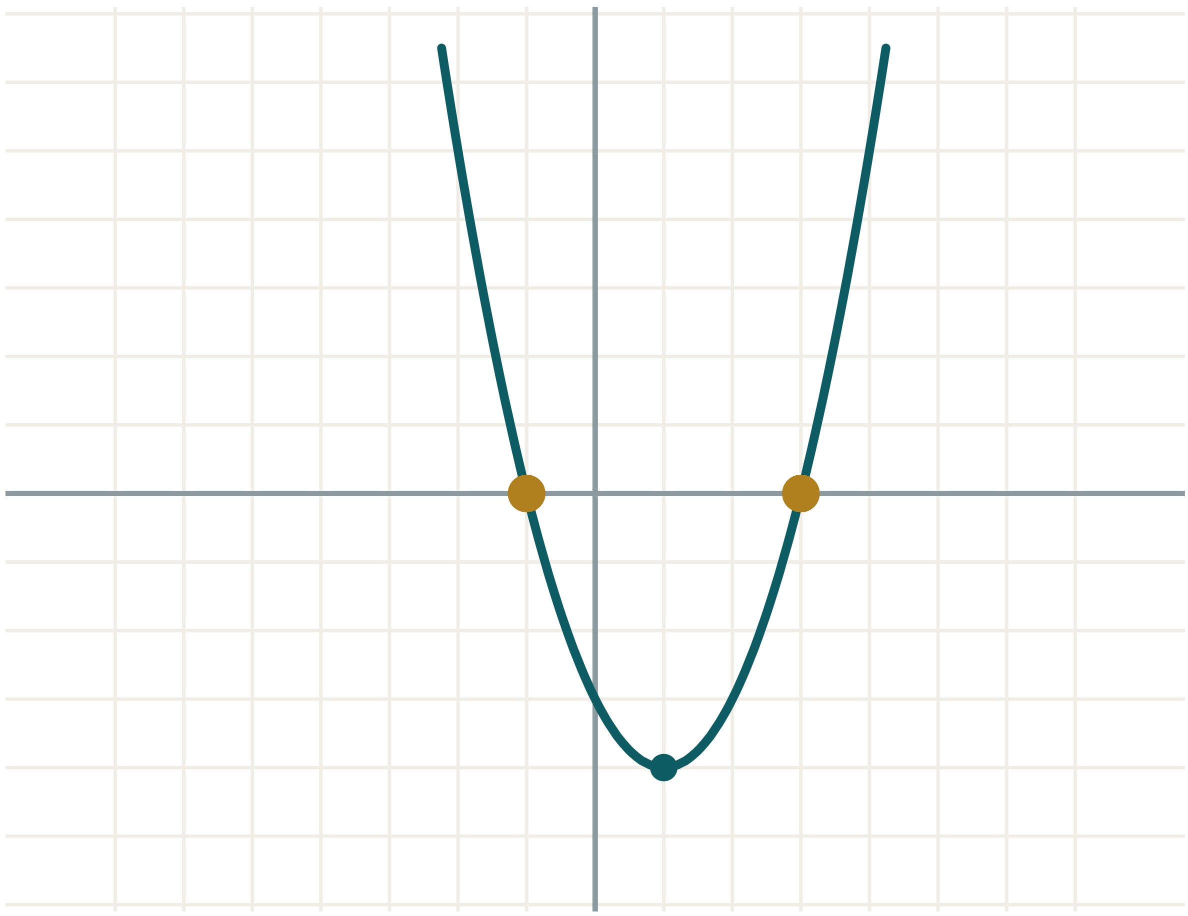
ANSWER

No roots.

## 03 Interactive model

Set b = 0 and vary c, then set c = 0 and vary b — the two incomplete shapes appear on the graph.

**Set b or c to zero**



a **1** | b **-2** | c **-3** | $D = b^2 - 4ac$ **16** | roots **two** | $x_1$ **-1** | $x_2$ **3** | $x_1 + x_2$ **2** | $x_1 \cdot x_2$ **-3**

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| Incomplete quadratic equation | Chala kvadrat tenglama | Неполное квадратное уравнение |
| Common factor | Umumiy ko‘paytuvchi | Общий множитель |
| Take the square root | Kvadrat ildiz chiqarish | Извлечь квадратный корень |
| Two opposite roots | Qarama-qarshi ildizlar | Противоположные корни |
| Lose a root | Ildizni yo‘qotish | Потерять корень |
| No real roots | Haqiqiy ildizga ega emas | Нет действительных корней |
| Zero root | Nol ildiz | Нулевой корень |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

## 05 Quick check

### Check yourself

$x^2 = 16$ gives:

- $x = 4$
- $x = -4$
- $x = \pm 4$
- $x = 8$

$x^2 + 9 = 0$ has:

- $x = \pm 3$
- $x = 3$
- no roots
- $x = 0$

$x^2 - 6x = 0$ gives:

- $x = 6$
- $x = 0, x = 6$
- $x = \pm 6$
- $x = 0$

Dividing $x^2 = 5x$ by $x$ is wrong because:

- it is too slow
- it loses the root x = 0
- $x^2$ is not divisible by x
- the answer becomes negative

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up 7 PROBLEMS

1. Solve $x^2 = 25$

ANSWER $x = \pm 5$

2. Solve $x^2 - 49 = 0$

ANSWER $x = \pm 7$

3. Solve $x^2 + 1 = 0$

ANSWER No roots.

4. Solve $x^2 = 0$

ANSWER $x = 0$

5. Solve $x^2 - 3x = 0$

ANSWER $x = 0, x = 3$

6. Solve $x^2 + 5x = 0$

ANSWER $x = 0, x = -5$

7. Solve $2x^2 = 8$

ANSWER $x = \pm 2$

## Medium · the standard question

7 PROBLEMS

1. Solve $5x^2 - 45 = 0$

ANSWER $x = \pm 3$

2. Solve $4x^2 + 7x = 0$

ANSWER $x = 0, x = -1.75$

3. Solve $3x^2 + 12 = 0$

ANSWER No roots.

4. Solve $3x^2 - 12x = 0$

ANSWER $x = 0, x = 4$

5. Solve $9x^2 - 4 = 0$

ANSWER $x = \pm\frac{2}{3}$

6. Solve $x^2 = 7x$

ANSWER $x = 0, x = 7$

7. Solve $2x^2 - 50 = 0$

ANSWER $x = \pm 5$

## Hard · stretch and reason

7 PROBLEMS

1. *Solve* $(x - 2)^2 = 9$

   ANSWER $x - 2 = \pm 3$: $x = 5$ or $x = -1$

2. *Solve* $4x^2 = 3x$

   ANSWER $x(4x - 3) = 0$: $x = 0$ or $x = 0.75$

3. *For which* $c$ *does* $x^2 + c = 0$ *have two roots?*

   ANSWER $c < 0$

4. *Solve* $2x^2 - 6 = 0$ *and give the answer exactly.*

   ANSWER $x^2 = 3$, so $x = \pm\sqrt{3}$

5. *Solve* $x(x + 3) = 4x$

   ANSWER $x^2 + 3x - 4x = 0 \Longrightarrow x(x - 1) = 0$: $x = 0, x = 1$

6. *The area of a square is 121 cm². Find its side, and say why only one answer makes sense.*

   ANSWER $x^2 = 121$ gives $x = \pm 11$; a length cannot be negative, so $11$ *cm*.

7. *Solve* $(2x - 1)^2 = 0$

   ANSWER $2x - 1 = 0$, so $x = 0.5$ — one root only.

## 07 Homework

### Homework — 6 problems

Algebra 8, Chapter III, pp. 129–133. Never divide by x.

1. *Solve* $x^2 - 64 = 0$
2. *Solve* $3x^2 - 27 = 0$
3. *Solve* $x^2 + 16 = 0$
4. *Solve* $5x^2 + 20x = 0$
5. *Solve* $x^2 = 9x$

6. *Solve* $(x + 1)^2 = 16$

# The quadratic formula and the discriminant

One formula that solves every quadratic — and one number inside it that says in advance how many roots there will be.

LESSON 66–67 | 2 × 40 MIN | ALGEBRA 8, CHAPTER III §22 | STAGE 9 · BEYOND

**LESSON CLOCK**

5' | 12' | 8' | 13' | 2'

| | |
|---|---|
| Warm-up | 5 min |
| Explanation | 12 min |
| Interactive | 8 min |
| Practice | 13 min |
| Homework | 2 min |

**LEARNING OBJECTIVES**

- Calculate the discriminant $D = b^2 - 4ac$.
- Say how many roots an equation has from the sign of $D$.
- Use the quadratic formula correctly, including with negative coefficients.
- Choose between factorising and the formula.

**TEXTBOOK**

| | |
|---|---|
| National | Chapter III, pp. 134–140 |
| Cambridge | Extension beyond Stage 9 |
| Workbook | – |

## 01 Explanation

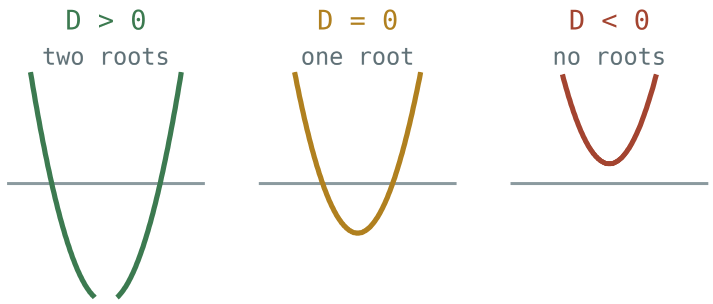
### The discriminant

DEFINITION

For $ax^2 + bx + c = 0$ the **discriminant** is

$$D = b^2 - 4ac$$

| D | ROOTS | THE PARABOLA |
|---|---|---|
| $D > 0$ | two different roots | crosses the x-axis twice |
| $D = 0$ | one root (repeated) | touches the axis |
| $D < 0$ | no real roots | misses the axis |




*The sign of the discriminant decides whether the parabola crosses, touches or misses the x-axis.*

Work out $D$ **first**, every time. If it is negative you can stop — there is nothing to find, and you have saved yourself a page of arithmetic.

## The formula

THE QUADRATIC FORMULA

$$x = \frac{-b \pm \sqrt{D}}{2a}, \quad \text{where } D = b^2 - 4ac$$

It comes from completing the square on the general equation, and it works for every quadratic whatever the coefficients.

THREE PLACES WHERE MARKS ARE LOST

1. $-b$ when $b$ is already negative: for $b = -5$, $-b = +5$.
2. $-4ac$ when $c$ is negative: $-4 \cdot 1 \cdot (-3) = +12$.
3. The whole of $-b \pm \sqrt{D}$ is divided by $2a$ — not just the root.

## Which method?

- **Incomplete** ($b = 0$ or $c = 0$) → the short methods of the last lesson.
- **Factorises easily** (small whole-number roots) → factorise; it is quicker.
- **Anything else** → the formula.

A useful check: if $D$ is a perfect square, the equation would have factorised. If it is not, the roots are irrational and the formula was the only route.

## 02 Worked examples

EXAMPLE 1 **Solve $2x^2 - 7x + 3 = 0$**

1. $a = 2, b = -7, c = 3$

   Read off the coefficients with their signs.

2. $D = (-7)^2 - 4 \cdot 2 \cdot 3 = 49 - 24 = 25$

   $D > 0$, so two roots.

3. $$x = \frac{7 \pm \sqrt{25}}{4} = \frac{7 \pm 5}{4}$$

   $-b = +7$

4. $x = 3$ or $x = 0.5$

   $\frac{12}{4}$ and $\frac{2}{4}$

ANSWER

$x = 3, x = 0.5$

EXAMPLE 2 ***Solve $x^2 + 4x + 4 = 0$***

1. $a = 1, b = 4, c = 4$
2. $D = 16 - 16 = 0$

   One repeated root.
3. $x = \frac{-4}{2} = -2$
4. $(x + 2)^2 = 0$

   The same answer by factorising.

ANSWER

$x = -2$

EXAMPLE 3 ***Solve $x^2 - 2x + 5 = 0$***

1. $a = 1, b = -2, c = 5$
2. $D = 4 - 20 = -16$

   $D < 0$.
3. No real roots — stop here.

   The parabola lies entirely above the x-axis.

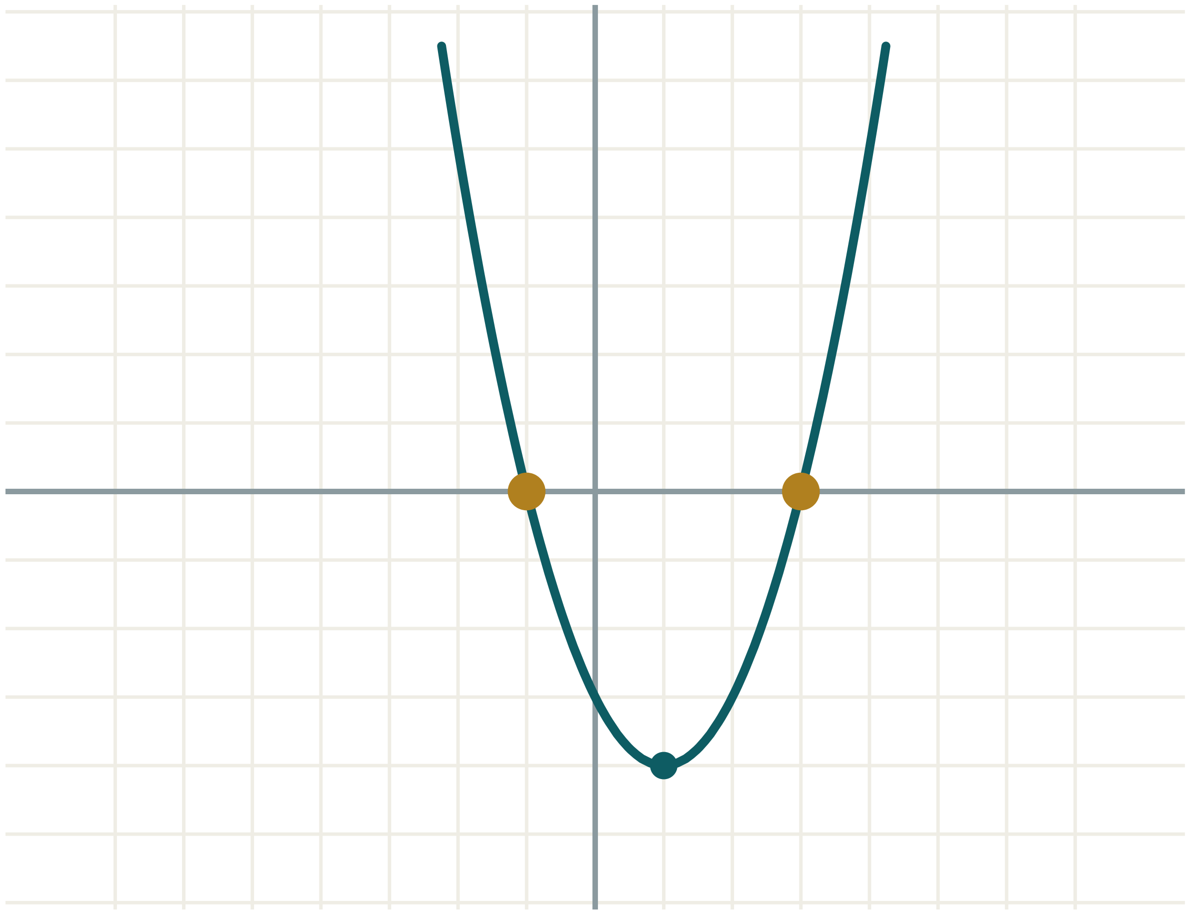
ANSWER

No real roots.

## 03 Interactive model

Move c downwards and watch D pass through zero as the parabola touches then crosses the axis.

**D decides everything**



a **1** | b **-2** | c **-3** | $D = b^2 - 4ac$ **16** | roots **two** | $x_1$ **-1** | $x_2$ **3** | $x_1 + x_2$ **2** | $x_1 \cdot x_2$ **-3**

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| Discriminant | Diskriminant | Дискриминант |
| Quadratic formula | Kvadrat tenglama formulasi | Формула корней квадратного уравнения |
| Two distinct roots | Ikkita har xil ildiz | Два различных корня |
| Repeated root | Karrali ildiz | Кратный корень |
| No real roots | Haqiqiy ildiz yo‘q | Нет действительных корней |
| Substitute | O‘rniga qo‘yish | Подставить |
| Simplify the surd | Ildizni soddalashtirish | Упростить корень |
| Exact answer | Aniq javob | Точный ответ |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

# 05 Quick check

### Check yourself

For $x^2 - 5x + 6 = 0$, D is:

- $1$
- $49$
- $-1$
- $25$

$D = 0$ means:

- no roots
- one root
- two roots
- infinitely many

For $2x^2 + 3x + 5 = 0$, D is:

- $49$
- $-31$
- $9$
- $31$

In the formula, when $b = -6$ the numerator starts with:

- $-6$
- $+6$
- $6^2$
- $-36$

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up

7 PROBLEMS

1. *Find D for* $x^2 + 3x + 2 = 0$

   ANSWER $1$

2. *Find D for* $x^2 - 4x + 4 = 0$

   ANSWER $0$

3. *Find D for* $x^2 + x + 1 = 0$

   ANSWER $-3$

4. *How many roots when* $D = 9$*?*

   ANSWER Two.

5. *How many roots when* $D = 0$*?*

   ANSWER One.

6. *How many roots when* $D = -5$*?*

   ANSWER None.

7. *Solve* $x^2 + 3x + 2 = 0$

   ANSWER $x = -1, x = -2$

## Medium · the standard question

7 PROBLEMS

1. Solve $2x^2 - 7x + 3 = 0$

ANSWER $x = 3, x = 0.5$

2. Solve $x^2 + 4x + 4 = 0$

ANSWER $x = -2$

3. Solve $x^2 - 2x + 5 = 0$

ANSWER No real roots.

4. Solve $3x^2 - 5x - 2 = 0$

ANSWER $D = 49$: $x = 2, x = -\frac{1}{3}$

5. Solve $x^2 - 6x + 9 = 0$

ANSWER $x = 3$

6. Solve $2x^2 + 5x - 3 = 0$

ANSWER $D = 49$: $x = 0.5, x = -3$

7. Solve $x^2 + 2x - 8 = 0$

ANSWER $x = 2, x = -4$

## Hard · stretch and reason

7 PROBLEMS

1. *Solve $x^2 - 4x + 1 = 0$ exactly*

   ANSWER $D = 12$: $x = 2 \pm \sqrt{3}$

2. *Solve $2x^2 - 4x - 1 = 0$ exactly*

   ANSWER $D = 24$: $x = \dfrac{2 \pm \sqrt{6}}{2}$

3. *For which k does $x^2 + kx + 9 = 0$ have exactly one root?*

   ANSWER $k^2 = 36$, so $k = \pm 6$

4. *For which k does $x^2 - 6x + k = 0$ have two roots?*

   ANSWER $36 - 4k > 0$, so $k < 9$

5. *For which k does $x^2 + 4x + k = 0$ have no roots?*

   ANSWER $16 - 4k < 0$, so $k > 4$

6. *Solve $5x^2 - 2x - 3 = 0$*

   ANSWER $D = 64$: $x = 1, x = -0.6$

7. *Show that $x^2 + x + 1 = 0$ has no real roots, without the formula.*

   ANSWER $x^2 + x + 1 = (x + 0.5)^2 + 0.75 \geq 0.75 > 0$ for every $x$.

## 07 Homework

### Homework — 6 problems

Algebra 8, Chapter III, pp. 134–140. Find D before anything else.

1. *Find D and solve $x^2 - 7x + 10 = 0$*
2. *Find D and solve $3x^2 + 2x - 1 = 0$*
3. *Find D and solve $x^2 + 2x + 5 = 0$*
4. *Solve $x^2 - 8x + 16 = 0$*
5. *Solve $x^2 - 6x + 2 = 0$ exactly*

6. *For which k does* $x^2 + kx + 4 = 0$ *have exactly one root?*

Home / Grade 8 / Algebra / Quarter 3

# Vieta's theorem and factorising a quadratic trinomial

The sum and the product of the roots, read straight off the coefficients — a check, a shortcut and a factorising method in one.

LESSON 68–69 | 2 × 40 MIN | ALGEBRA 8, CHAPTER III §23 | STAGE 9 · 2.4

LESSON CLOCK


| | |
|---|---|
| Warm-up | 5 min |
| Explanation | 12 min |
| Interactive | 8 min |
| Practice | 13 min |
| Homework | 2 min |

LEARNING OBJECTIVES

- State and use Vieta's theorem.
- Check a pair of roots without substituting.
- Find roots by inspection for a reduced quadratic.
- Factorise $ax^2 + bx + c$ using its roots.

TEXTBOOK

| | |
|---|---|
| National | Chapter III, pp. 141–146 |
| Cambridge | Learner's Book pp. 36–39 |
| Workbook | Workbook 2.4 |

## 01 Explanation

### The theorem

VIETA'S THEOREM

If $x_1$ and $x_2$ are the roots of $ax^2 + bx + c = 0$, then

$$x_1 + x_2 = -\frac{b}{a} \quad \text{and} \quad x_1 \cdot x_2 = \frac{c}{a}$$

For a **reduced** equation $x^2 + px + q = 0$ (with $a = 1$) it is even simpler:

$$x_1 + x_2 = -p \quad \text{and} \quad x_1 \cdot x_2 = q$$

Check it on $x^2 - 5x + 6 = 0$, whose roots are 2 and 3: $2 + 3 = 5 = -(-5)$ ✓ and $2 \cdot 3 = 6$ ✓.

### Three uses

**1 · As a check.** After solving, add and multiply your two roots. If they do not give $-\frac{b}{a}$ and $\frac{c}{a}$, go back — this takes five seconds and catches almost every arithmetic slip.

**2 · To find roots by inspection.** For $x^2 - 7x + 12 = 0$, look for two numbers with sum 7 and product 12: 3 and 4. No formula needed.

**3 · To find an unknown coefficient.** If one root of $x^2 + bx - 10 = 0$ is 5, then the product of the roots is $-10$, so the other root is $-2$, and the sum $5 + (-2) = 3 = -b$ gives $b = -3$.

## Factorising the trinomial

THEOREM

If $x_1$ and $x_2$ are the roots of $ax^2 + bx + c = 0$, then

$$ax^2 + bx + c = a(x - x_1)(x - x_2)$$

So factorising a quadratic trinomial and solving the equation are the same job. Find the roots, then write the brackets:

> $2x^2 - 7x + 3$: roots $3$ and $0.5$, so $= 2(x - 3)(x - 0.5) = (x - 3)(2x - 1)$

DO NOT FORGET THE A

$2x^2 - 7x + 3$ is not $(x - 3)(x - 0.5)$ — that expands to $x^2 - 3.5x + 1.5$. The leading coefficient must be carried through.

If $D < 0$ there are no roots, and the trinomial **cannot** be factorised over the real numbers.

## 02 Worked examples

EXAMPLE 1 ***Solve $x^2 - 9x + 20 = 0$ by inspection.***

1. Sum of roots $= 9$, product $= 20$.

   Vieta for a reduced equation.

2. Two numbers with product 20: 1·20, 2·10, 4·5.

3. $4 + 5 = 9$ ✓

4. $x = 4$ and $x = 5$

ANSWER

$x = 4, x = 5$

EXAMPLE 2 ***One root of $x^2 + bx - 21 = 0$ is 7. Find the other root and b.***

1. $x_1 \cdot x_2 = -21$

   The product equals the free term.

2. $7 \cdot x_2 = -21$, so $x_2 = -3$

3. $x_1 + x_2 = 7 + (-3) = 4$

4. $-b = 4$, so $b = -4$

ANSWER

$x_2 = -3, b = -4$

EXAMPLE 3 ***Factorise $3x^2 - 5x - 2$***

1. $D = 25 + 24 = 49$

2. $x = \dfrac{5 \pm 7}{6}$

   The formula.

3. $x_1 = 2, x_2 = -\dfrac{1}{3}$

4. $3(x - 2)(x + \dfrac{1}{3}) = (x - 2)(3x + 1)$

   Multiply the 3 into the second bracket.

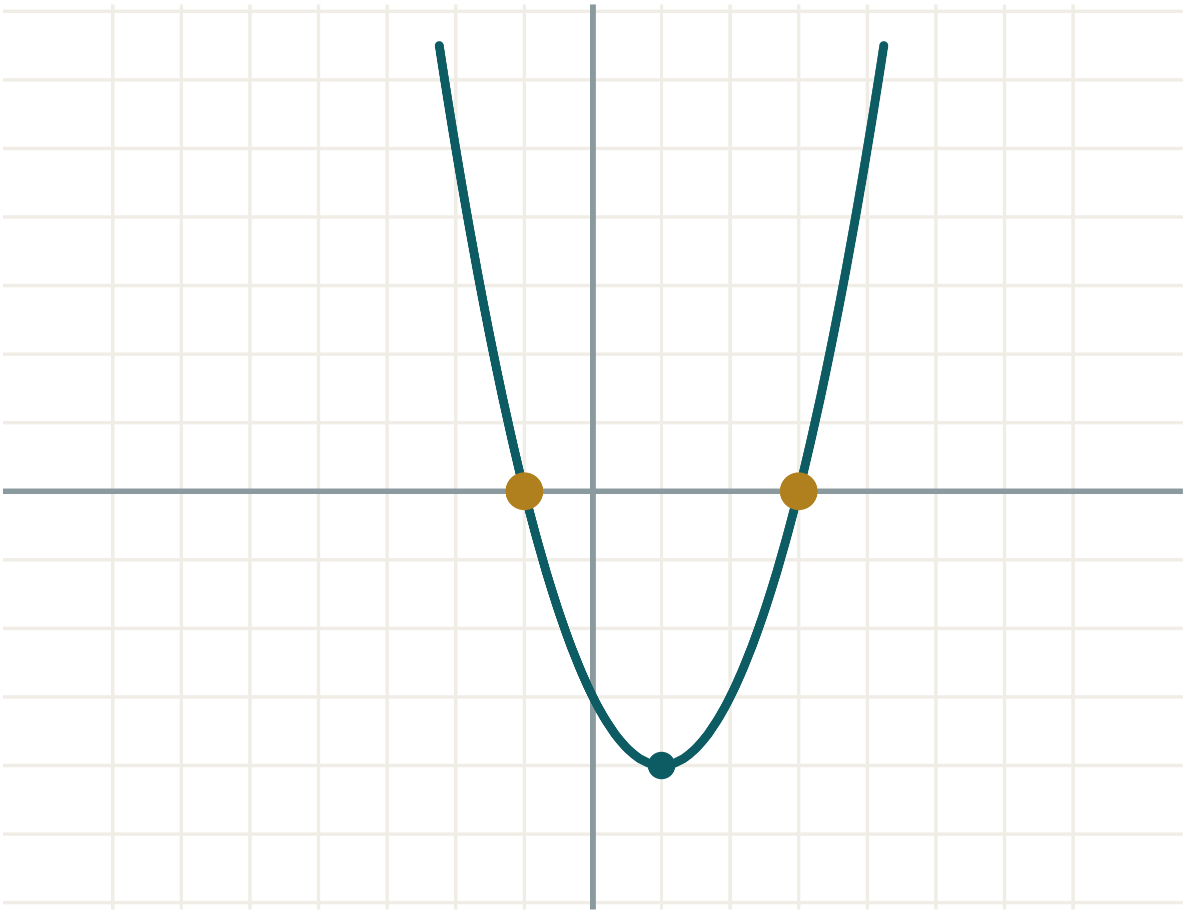
ANSWER

$(x - 2)(3x + 1)$

## 03 Interactive model

Read $x_1 + x_2$ and $x_1 \cdot x_2$ from the model and compare with $-b/a$ and $c/a$.

**Sum and product of the roots**



a **1** | b **-2** | c **-3** | D = $b^2 - 4ac$ **16** | roots **two** | $x_1$ **-1** | $x_2$ **3** | $x_1 + x_2$ **2** | $x_1 \cdot x_2$ **-3**

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| Vieta’s theorem | Viyet teoremasi | Теорема Виета |
| Sum of the roots | Ildizlar yig‘indisi | Сумма корней |
| Product of the roots | Ildizlar ko‘paytmasi | Произведение корней |
| Reduced quadratic equation | Keltirilgan kvadrat tenglama | Приведённое квадратное уравнение |
| Quadratic trinomial | Kvadrat uchhad | Квадратный трёхчлен |
| By inspection | Tanlash yo‘li bilan | Подбором |
| Linear factors | Chiziqli ko‘paytuvchilar | Линейные множители |
| Converse theorem | Teskari teorema | Обратная теорема |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

# 05 Quick check

### Check yourself

For $x^2 - 7x + 12 = 0$, the sum of the roots is:

- $-7$
- $7$
- $12$
- $-12$

For the same equation, the product is:

- $7$
- $12$
- $-12$
- $-7$

For $2x^2 + 6x - 8 = 0$, the sum of the roots is:

- $-6$
- $-3$
- $3$
- $6$

$x^2 - 5x + 6$ factorises to:

- $(x - 2)(x - 3)$
- $(x + 2)(x + 3)$
- $(x - 1)(x - 6)$
- it does not factorise

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up 7 PROBLEMS

1. *Find the sum of the roots of* $x^2 - 4x + 3 = 0$

ANSWER $4$

2. *Find the product of the roots of* $x^2 - 4x + 3 = 0$

ANSWER $3$

3. *Solve* $x^2 - 5x + 6 = 0$ *by inspection*

ANSWER $x = 2, x = 3$

4. *Solve* $x^2 - 7x + 12 = 0$ *by inspection*

ANSWER $x = 3, x = 4$

5. *Solve* $x^2 + 5x + 6 = 0$ *by inspection*

ANSWER $x = -2, x = -3$

6. *Factorise* $x^2 - 3x + 2$

ANSWER $(x - 1)(x - 2)$

7. *Factorise* $x^2 - 16$

ANSWER $(x - 4)(x + 4)$

## Medium · the standard question

7 PROBLEMS

1. *Solve $x^2 - 9x + 20 = 0$ by inspection*

   ANSWER $x = 4, x = 5$

2. *Solve $x^2 + x - 12 = 0$ by inspection*

   ANSWER $x = 3, x = -4$

3. *One root of $x^2 + bx - 21 = 0$ is 7. Find the other root and b.*

   ANSWER $x_2 = -3, b = -4$

4. *For $2x^2 + 6x - 8 = 0$, find the sum and product of the roots.*

   ANSWER $-3$ and $-4$

5. *Factorise $x^2 - 2x - 15$*

   ANSWER $(x - 5)(x + 3)$

6. *Factorise $3x^2 - 5x - 2$*

   ANSWER $(x - 2)(3x + 1)$

7. *Write a reduced quadratic whose roots are 6 and −2.*

   ANSWER $x^2 - 4x - 12 = 0$

## Hard · stretch and reason

7 PROBLEMS

1. *The roots of $x^2 + px + q = 0$ are 3 and $-5$. Find p and q.*

   ANSWER $p = 2, q = -15$

2. *One root of $2x^2 - 9x + c = 0$ is 4. Find c and the other root.*

   ANSWER Sum $= 4.5$, so the other root is $0.5$ and $c = 2 \cdot 4 \cdot 0.5 = 4$.

3. *Factorise $2x^2 + 5x - 12$*

   ANSWER $(x + 4)(2x - 3)$

4. *The roots of $x^2 - 6x + q = 0$ are equal. Find q and the root.*

   ANSWER $D = 0$ gives $q = 9$ and $x = 3$.

5. *For $x^2 - 5x + 6 = 0$, find $x_1^2 + x_2^2$ without solving.*

   ANSWER $(x_1 + x_2)^2 - 2x_1x_2 = 25 - 12 = 13$

6. *For $x^2 + 3x - 10 = 0$, find $\frac{1}{x_1} + \frac{1}{x_2}$ without solving.*

   ANSWER $\frac{x_1 + x_2}{x_1x_2} = \frac{-3}{-10} = 0.3$

7. *Can $x^2 + x + 3$ be factorised into linear factors? Explain.*

   ANSWER No — $D = 1 - 12 = -11 < 0$, so there are no real roots.

## 07 Homework

### Homework — 6 problems

Algebra 8, Chapter III, pp. 141–146. Use Vieta as a check on every answer.

1. *Find the sum and product of the roots of $x^2 - 11x + 30 = 0$*
2. *Solve $x^2 - 8x + 15 = 0$ by inspection*
3. *One root of $x^2 + bx - 24 = 0$ is 6. Find the other root and b.*
4. *Factorise $x^2 + 4x - 21$*

5. *Factorise* $2x^2 - 7x + 3$

6. *Write a reduced quadratic equation with roots* $-1$ *and* $8$

# Biquadratic equations and equations reducible to quadratics

One substitution turns a degree-four equation, or a fractional one, into a quadratic you can already solve.

LESSON 70–71 | 2 × 40 MIN | ALGEBRA 8, CHAPTER III §24 | STAGE 9 · BEYOND

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

| | |
|---|---|
| Warm-up | 5 min |
| Explanation | 12 min |
| Interactive | 8 min |
| Practice | 13 min |
| Homework | 2 min |

LEARNING OBJECTIVES

- Solve a biquadratic equation by the substitution $t = x^2$.
- Reject negative values of $t$ and explain why.
- Solve a fractional-rational equation that reduces to a quadratic.
- Check for values excluded by the denominator.

TEXTBOOK

| | |
|---|---|
| National | Chapter III, pp. 147–152 |
| Cambridge | Extension beyond Stage 9 |
| Workbook | – |

## 01 Explanation

### Biquadratic equations

DEFINITION

An equation of the form $ax^4 + bx^2 + c = 0$ is called **biquadratic**. Only even powers appear.

Put $t = x^2$. Since $x^4 = (x^2)^2 = t^2$, the equation becomes

$$at^2 + bt + c = 0$$

— an ordinary quadratic. Solve it, then **return to $x$** by solving $x^2 = t$ for each value of $t$ found.

THE STEP EVERYONE FORGETS

$t = x^2$ is a square, so $t \geq 0$. Any **negative** value of $t$ must be **rejected** — it gives no real $x$. And every positive $t$ gives **two** values of $x$, namely $\pm\sqrt{t}$.

So a biquadratic equation can have 4, 3, 2, 1 or 0 real roots — count carefully.

## A worked pattern

$$x^4 - 13x^2 + 36 = 0$$

$$t^2 - 13t + 36 = 0 \ \ \text{(with } t = x^2\text{)}$$

$$t = 4 \text{ or } t = 9 \ \ \text{(sum 13, product 36)}$$

$$x^2 = 4 \Longrightarrow x = \pm 2 \quad x^2 = 9 \Longrightarrow x = \pm 3$$

Four roots: $-3, -2, 2, 3$.

## Fractional equations that reduce to quadratics

Multiply through by the common denominator, then solve the quadratic that appears — but list the excluded values **first**.

$$\frac{x}{x-2} + \frac{4}{x} = 3, \quad x \neq 0, x \neq 2$$

Multiplying by $x(x - 2)$:

$$x^2 + 4(x - 2) = 3x(x - 2)$$

$$x^2 + 4x - 8 = 3x^2 - 6x \Longrightarrow 2x^2 - 10x + 8 = 0 \Longrightarrow x^2 - 5x + 4 = 0$$

$$x = 1 \text{ or } x = 4 \text{ — both permissible ✓}$$

CHECK AGAINST THE EXCLUDED LIST

If a root turns out to be an excluded value it is an **extraneous root** and must be thrown away. Multiplying by an expression that could be zero is exactly what creates them.

# 02 Worked examples

## EXAMPLE 1 Solve $x^4 - 5x^2 + 4 = 0$

1. $t = x^2$: $t^2 - 5t + 4 = 0$

   The substitution.

2. $t = 1$ or $t = 4$

   Sum 5, product 4.

3. $x^2 = 1 \Longrightarrow x = \pm 1$

4. $x^2 = 4 \Longrightarrow x = \pm 2$

   Four roots in all.

ANSWER

$x = \pm 1,\ x = \pm 2$

## EXAMPLE 2 Solve $x^4 + 3x^2 - 4 = 0$

1. $t^2 + 3t - 4 = 0$

2. $t = 1$ or $t = -4$

3. $t = -4$ is rejected — a square cannot be negative.

   This is the key step.

4. $x^2 = 1 \Longrightarrow x = \pm 1$

   Only two roots.

ANSWER

$x = \pm 1$

EXAMPLE 3 ***Solve*** $\frac{6}{x} = x - 1$

1. $x \neq 0$

   Note the exclusion first.

2. $6 = x(x - 1)$

   Multiply by x.

3. $x^2 - x - 6 = 0$

4. $x = 3$ or $x = -2$

   Neither is 0, so both stand.

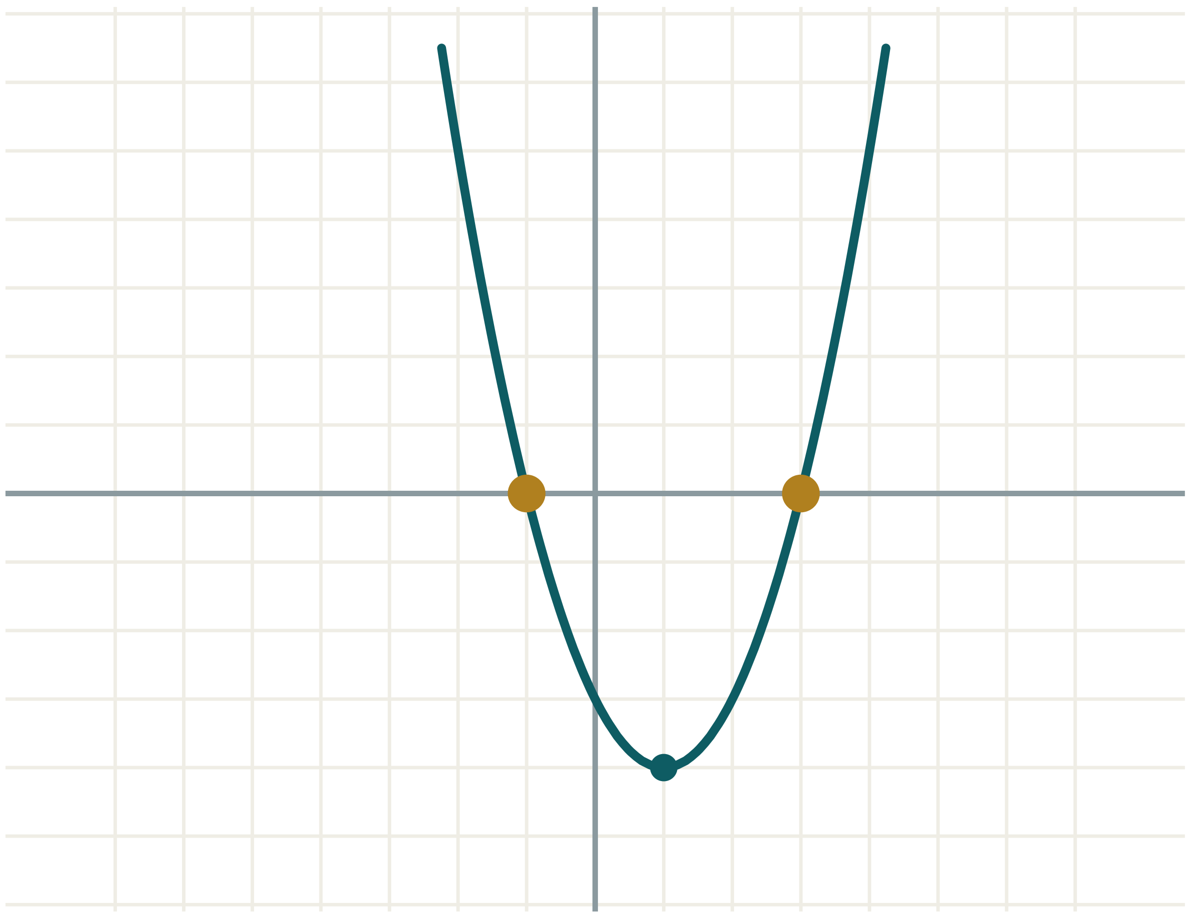
ANSWER

$x = 3, x = -2$

## 03 Interactive model

Use the model on the quadratic in t, then ask the class how many x-values each t produces.

**The quadratic in t**



a **1** | b **-2** | c **-3** | D = $b^2 - 4ac$ **16** | roots **two** | $x_1$ **-1** | $x_2$ **3** | $x_1 + x_2$ **2** | $x_1 \cdot x_2$ **-3**

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O'ZBEKCHA | РУССКИЙ |
|---|---|---|
| Biquadratic equation | Bikvadrat tenglama | Биквадратное уравнение |
| Substitution | Almashtirish | Замена переменной |
| New variable | Yangi oʻzgaruvchi | Новая переменная |
| Reject a value | Qiymatni rad etish | Отбросить значение |
| Fractional-rational equation | Kasr-ratsional tenglama | Дробно-рациональное уравнение |
| Extraneous root | Chet ildiz | Посторонний корень |
| Return to x | x ga qaytish | Вернуться к x |
| Degree of an equation | Tenglamaning darajasi | Степень уравнения |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

## 05 Quick check

### Check yourself

For a biquadratic the substitution is:

- $t = x$
- $t = x^2$
- $t = x^4$
- $t = \sqrt{x}$

If $t = -9$ comes out of the substitution, you should:

- take $x = \pm 3$
- reject it
- take $x = -3$
- start again

$x^4 - 13x^2 + 36 = 0$ has how many real roots?

- $2$
- $3$
- $4$
- $0$

Before multiplying a fractional equation by its denominator you must:

- square both sides
- list the excluded values
- factorise the numerator
- nothing

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up  7 PROBLEMS

1. *Solve* $x^4 - 5x^2 + 4 = 0$

   ANSWER $x = \pm 1, \pm 2$

2. *Solve* $x^4 - 10x^2 + 9 = 0$

   ANSWER $x = \pm 1, \pm 3$

3. *Solve* $x^4 - 16 = 0$

   ANSWER $x = \pm 2$

4. *Solve* $x^4 - 9x^2 = 0$

   ANSWER $x = 0, \pm 3$

5. *If* $t = x^2$ *and* $t = 25$, *find* $x$.

   ANSWER $x = \pm 5$

6. *If* $t = x^2$ *and* $t = -1$, *find* $x$.

   ANSWER No real x.

7. *Solve* $\frac{4}{x} = 2$

   ANSWER $x = 2$

## Medium · the standard question

7 PROBLEMS

1. Solve $x^4 + 3x^2 - 4 = 0$

ANSWER $x = \pm 1$

2. Solve $x^4 - 13x^2 + 36 = 0$

ANSWER $x = \pm 2, \pm 3$

3. Solve $x^4 + 5x^2 + 4 = 0$

ANSWER No real roots — both t values are negative.

4. Solve $\dfrac{6}{x} = x - 1$

ANSWER $x = 3, x = -2$

5. Solve $\dfrac{12}{x} + x = 7$

ANSWER $x = 3, x = 4$

6. Solve $x^4 - 8x^2 + 16 = 0$

ANSWER $t = 4$ twice, so $x = \pm 2$

7. Solve $\dfrac{1}{x} + \dfrac{1}{x+1} = \dfrac{5}{6}$

ANSWER $x = 2$ or $x = -3$

## Hard · stretch and reason

7 PROBLEMS

1. Solve $4x^4 - 17x^2 + 4 = 0$

ANSWER $t = 4$ or $t = 0.25$: $x = \pm 2, \pm 0.5$

2. Solve $\dfrac{x}{x-2} + \dfrac{4}{x} = 3$

ANSWER $x = 1, x = 4$

3. Solve $\dfrac{x+1}{x-1} - \dfrac{x-1}{x+1} = 1$

ANSWER $4x = x^2 - 1$: $x = 2 \pm \sqrt{5}$

4. Solve $(x^2 - 1)^2 - 5(x^2 - 1) + 4 = 0$

ANSWER Let $t = x^2 - 1$: $t = 1$ or $4$, so $x = \pm\sqrt{2}, \pm\sqrt{5}$

5. For which k does $x^4 + kx^2 + 4 = 0$ have four real roots?

ANSWER Need two positive t-roots: $k^2 > 16$ and $-k > 0$, so $k < -4$.

6. Solve $\dfrac{2}{x-3} = \dfrac{x}{x-3} - 1$

ANSWER $x \neq 3$; $2 = x - (x - 3) = 3$ — false, so no solutions.

7. Solve $\dfrac{x^2-4}{x-2} = 5$

ANSWER $x \neq 2$; $x + 2 = 5$, so $x = 3$.

## 07 Homework

### Homework — 6 problems

Algebra 8, Chapter III, pp. 147–152. Reject negative t values and say so in writing.

1. Solve $x^4 - 26x^2 + 25 = 0$
2. Solve $x^4 + 2x^2 - 3 = 0$
3. Solve $x^4 + 4x^2 + 3 = 0$

4. *Solve* $\frac{8}{x} = x + 2$

5. *Solve* $\frac{20}{x} - x = 1$

6. *Solve* $x^4 - 4x^2 = 0$

# Solving problems with quadratic equations

Turning words into a quadratic — and deciding which of the two roots the problem actually allows.

LESSON 72–73 | 2 × 40 MIN | ALGEBRA 8, CHAPTER III §25 | STAGE 9 · 4.1

LESSON CLOCK

5' 12' 8' 13' 2'

| | |
|---|---|
| Warm-up | 5 min |
| Explanation | 12 min |
| Interactive | 8 min |
| Practice | 13 min |
| Homework | 2 min |

LEARNING OBJECTIVES

- Choose the unknown and write the equation.
- Solve and interpret both roots in the context.
- Reject a root that has no meaning in the problem.
- Handle the standard types: area, consecutive numbers, speed and work.

TEXTBOOK

| | |
|---|---|
| National | Chapter III, pp. 153–158 |
| Cambridge | Learner’s Book pp. 84–90 |
| Workbook | Workbook 4.1 |

## 01 Explanation

### The method

1. **Name the unknown.** Write “Let $x$ be …” and say what unit it is in.
2. **Express everything else** in terms of $x$.
3. **Write the equation** from the sentence that has not been used yet.
4. **Solve** it.
5. **Check both roots against the problem** and reject any that make no sense.
6. **Answer in words**, with units.

STEP 5 IS THE ONE THAT IS NEW

A quadratic gives two roots, but a length, a speed, a time or a number of people cannot be negative. Say explicitly which root you are rejecting and why — it carries a mark.

### The four standard types

**Area.** A rectangle is 3 m longer than it is wide and has area 40 m². Let the width be $x$; then $x(x + 3) = 40$, giving $x = 5$ or $x = -8$. A width cannot be negative, so the rectangle is $5\ m$ by $8\ m$.

**Consecutive numbers.** Two consecutive whole numbers have product 156. Let them be $x$ and $x + 1$: $x^2 + x - 156 = 0$, so $x = 12$ or $x = -13$. Both are valid here — 12 and 13, or $-13$ and $-12$.

**Speed.** Distance is $s = vt$, so a change of speed on a fixed distance gives a fractional equation that clears to a quadratic.

**Pythagoras.** A right triangle with one leg $x$ and the other $x + 7$ and hypotenuse 13: $x^2 + (x + 7)^2 = 169$.

## A worked speed problem

A cyclist covers 60 km. If the speed were 5 km/h greater, the journey would take one hour less. Find the speed.

Let the speed be $x$ km/h, $x > 0$. The times are $\frac{60}{x}$ and $\frac{60}{x+5}$, and the first is one hour longer:

$$\frac{60}{x} - \frac{60}{x+5} = 1$$

$$60(x + 5) - 60x = x(x + 5)$$

$$300 = x^2 + 5x \Longrightarrow x^2 + 5x - 300 = 0$$

$$x = 15 \text{ or } x = -20 \text{ — a speed cannot be negative}$$

The speed is $15\ km/h$.

# 02 Worked examples

**EXAMPLE 1** **A rectangle is 3 m longer than it is wide and has area 40 m². Find its dimensions.**

1. Let the width be $x$ m; the length is $x + 3$ m.

2. $x(x + 3) = 40$

   Area.

3. $x^2 + 3x - 40 = 0$

4. $x = 5$ or $x = -8$

5. A width cannot be negative, so $x = 5$.

   Reject the other root.

ANSWER

$5$ m by $8$ m

**EXAMPLE 2** **The product of two consecutive whole numbers is 156. Find them.**

1. Let them be $x$ and $x + 1$.

2. $x(x + 1) = 156$

3. $x^2 + x - 156 = 0$

4. $x = 12$ or $x = -13$

   Both give whole numbers here.

ANSWER

$12$ and $13$, or $-13$ and $-12$

EXAMPLE 3 **A cyclist covers 60 km. Riding 5 km/h faster would save an hour. Find the speed.**

1. Let the speed be $x$ km/h.

2. $$\frac{60}{x} - \frac{60}{x + 5} = 1$$
   The time saved.

3. $$x^2 + 5x - 300 = 0$$
   Multiply by $x(x + 5)$.

4. $x = 15$ or $x = -20$
   Reject the negative speed.

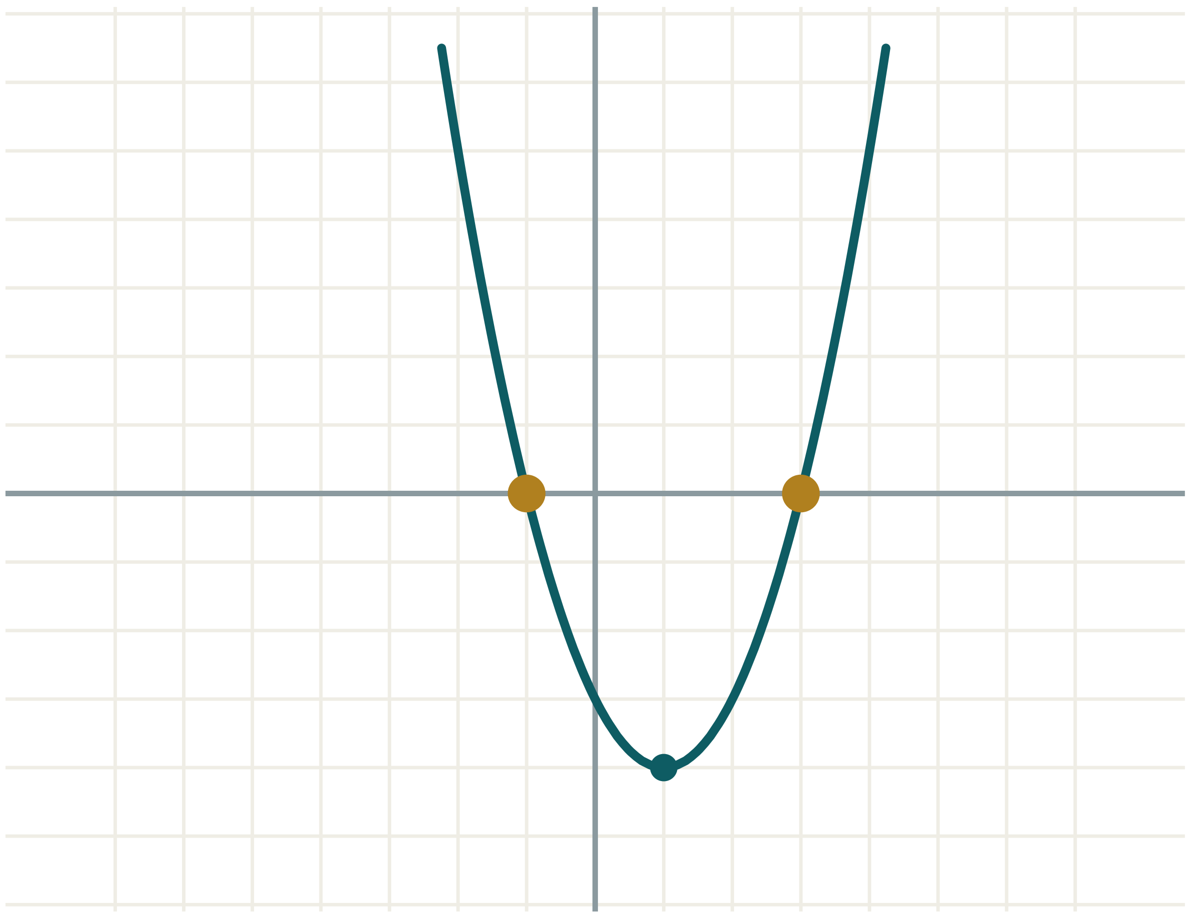
ANSWER

$15$ km/h

## 03 Interactive model

Write the equation on the board and let the model show which root the graph gives.

### The equation behind the problem



| | | | | | | | |
|---|---|---|---|---|---|---|---|
| a **1** | b **-2** | c **-3** | $D = b^2 - 4ac$ **16** | roots **two** | $x_1$ **-1** | $x_2$ **3** | $x_1 + x_2$ **2** |
| $x_1 \cdot x_2$ **-3** | | | | | | | |

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
| --- | --- | --- |
| Word problem | Matnli masala | Текстовая задача |
| Let x be… | x deb belgilaymiz | Пусть x — |
| Set up an equation | Tenglama tuzish | Составить уравнение |
| Reject a root | Ildizni rad etish | Отбросить корень |
| Consecutive numbers | Ketma-ket sonlar | Последовательные числа |
| Average speed | O‘rtacha tezlik | Средняя скорость |
| Dimensions | O‘lchamlar | Размеры |
| Interpret the answer | Javobni izohlash | Истолковать ответ |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

# 05 Quick check

## Check yourself

A rectangle has width $x$ and length $x + 4$ and area 45. The equation is:

- $2x + 4 = 45$
- $x(x + 4) = 45$
- $x + 4 = 45$
- $x^2 = 45$

A length comes out as $x = -6$ or $x = 4$. You should:

- take both
- take −6
- take 4 and reject −6
- start again

Two consecutive numbers are written as:

- $x$ and $2x$
- $x$ and $x + 1$
- $x$ and $x + 2$
- $x$ and $x^2$

Time equals:

- $distance \times speed$
- $\dfrac{\text{distance}}{\text{speed}}$
- $\dfrac{\text{speed}}{\text{distance}}$
- $distance + speed$

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up 7 PROBLEMS

1. *A square has area 81 cm². Find its side.*

   ANSWER $9$ cm

2. *A number squared is 49. Find it, given it is positive.*

   ANSWER $7$

3. *Two consecutive numbers have product 20. Find them.*

   ANSWER $4$ and $5$ (or $-5$ and $-4$)

4. *A rectangle is 2 m longer than wide, area 24 m². Write the equation.*

   ANSWER $x(x + 2) = 24$

5. *Solve that equation.*

   ANSWER $x = 4$, so $4$ m × $6$ m

6. *A number added to its square gives 12. Find the positive number.*

   ANSWER $3$

7. *The area of a square is 144. Find the perimeter.*

   ANSWER $48$

## Medium · the standard question

7 PROBLEMS

1. *A rectangle is 3 m longer than wide and has area 40 m². Find its dimensions.*

   ANSWER $5\ m \times 8\ m$

2. *The product of two consecutive whole numbers is 156. Find them.*

   ANSWER $12$ and $13$

3. *A right triangle has legs $x$ and $x + 7$ and hypotenuse 13. Find the legs.*

   ANSWER $5$ and $12$

4. *A number is 3 less than its square. Find both values.*

   ANSWER $x^2 - x - 3 = 0$: $x = \dfrac{1 \pm \sqrt{13}}{2}$

5. *The sum of a number and its reciprocal is 2.5. Find the number.*

   ANSWER $x = 2$ or $x = 0.5$

6. *A rectangle has perimeter 26 cm and area 40 cm². Find its sides.*

   ANSWER $5\ cm$ and $8\ cm$

7. *Two consecutive even numbers have product 168. Find them.*

   ANSWER $12$ and $14$

## Hard · stretch and reason

7 PROBLEMS

1. *A cyclist covers 60 km. Riding 5 km/h faster would save an hour. Find the speed.*

   ANSWER *15 km/h*

2. *A train covers 240 km. Going 20 km/h faster would save one hour. Find the speed.*

   ANSWER $x^2 + 20x - 4800 = 0$: *60 km/h*

3. *A rectangular garden 20 m by 15 m has a path of uniform width around it, doubling the area. Find the width.*

   ANSWER $(20 + 2w)(15 + 2w) = 600$: $w = 2.5$ *m*

4. *The sum of the squares of two consecutive whole numbers is 313. Find them.*

   ANSWER $x^2 + (x + 1)^2 = 313$: *12* and *13*

5. *A stone is thrown up and its height is $h = 20t - 5t^2$. When is it 15 m high?*

   ANSWER $5t^2 - 20t + 15 = 0$: at $t = 1$ s going up and $t = 3$ s coming down.

6. *Two workers together finish a job in 6 days. Alone, one takes 5 days longer than the other. Find both times.*

   ANSWER $\frac{1}{x} + \frac{1}{x+5} = \frac{1}{6}$: *10* and *15* days.

7. *The area of a triangle is 30 cm² and its height is 7 cm more than its base. Find the base.*

   ANSWER $\frac{1}{2}x(x + 7) = 30$: $x = 5$ *cm*

## 07 Homework

### Homework — 6 problems

Algebra 8, Chapter III, pp. 153–158. Always say which root you reject and why.

1. *A rectangle is 5 cm longer than wide and has area 84 cm². Find its dimensions.*
2. *Two consecutive whole numbers have product 342. Find them.*
3. *A right triangle has legs $x$ and $x + 2$ and hypotenuse 10. Find the legs.*
4. *A car covers 180 km. Going 10 km/h faster would save half an hour. Find the speed.*

5. *The sum of a number and its square is 56. Find both values.*

6. *A rectangle has perimeter 34 m and area 60 m². Find its sides.*

# Compound percentages, simultaneous equations and quadratic graphs

The national "practical and interdisciplinary" lessons, carrying three Cambridge Stage 9 topics the programme does not otherwise reach.

LESSON 74–76 | 3 × 40 MIN | ALGEBRA 8, PRACTICAL PROBLEMS

STAGE 9 · 3.3, 4.2, 10.2 [CAMBRIDGE INSERT]

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

| | |
|---|---|
| Warm-up | 5 min |
| Explanation | 12 min |
| Interactive | 8 min |
| Practice | 13 min |
| Homework | 2 min |

LEARNING OBJECTIVES

- Calculate repeated percentage change with a multiplier.
- Solve a pair of simultaneous linear equations by substitution and by elimination.
- Plot the graph of a quadratic and read its roots and vertex.
- Apply all three to practical situations.

TEXTBOOK

| | |
|---|---|
| National | pp. 159–164 |
| Cambridge | Learner's Book pp. 66–73, 91–95, 218–224 |
| Workbook | Workbook 3.3, 4.2, 10.2 |

## 01 Explanation

### Lesson 74 — compound percentage change

THE MULTIPLIER

An increase of $p\%$ multiplies by $1 + \frac{p}{100}$; a decrease multiplies by $1 - \frac{p}{100}$. For $n$ repetitions, raise the multiplier to the power $n$:

$$final = initial \times (multiplier)^n$$

A salary of 2 000 000 so'm rising 8% a year for 3 years: $2\ 000\ 000 \times 1.08^3 \approx 2\ 519\ 424$.

NOT THE SAME AS ADDING THE PERCENTAGES

> Three years at 8% is **not** 24%. It is $1.08^3 \approx 1.2597$, that is $25.97\%$ — the extra comes from the growth compounding on itself.

A 20% rise followed by a 20% fall is $1.2 \times 0.8 = 0.96$ — a 4% **loss**, not a return to the start. This is the single most useful fact in the topic.

## Lesson 75 — simultaneous equations

**Substitution.** Make one unknown the subject of one equation and put it into the other.

> $y = 2x + 1$ and $3x + y = 11 \Longrightarrow 3x + 2x + 1 = 11 \Longrightarrow x = 2, y = 5$

**Elimination.** Add or subtract the equations so that one unknown disappears — multiplying one or both first if you need matching coefficients.

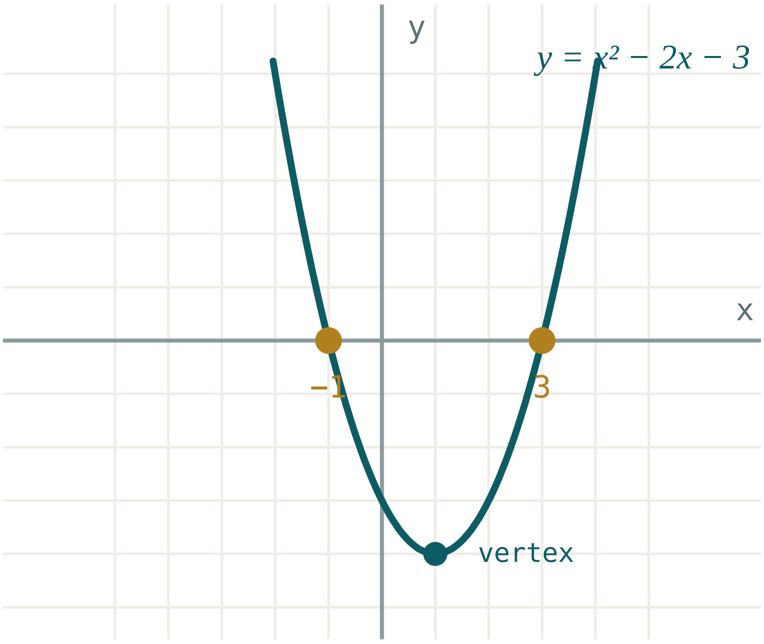
> $2x + 3y = 12$ and $2x - y = 4$: subtract $\Longrightarrow 4y = 8 \Longrightarrow y = 2, x = 3$

**Graphically** the solution is where the two lines cross. Parallel lines mean no solution; the same line twice means infinitely many.

Always **check in both equations** — a pair that satisfies only one is not a solution.

## Lesson 76 — the graph of a quadratic

To draw $y = ax^2 + bx + c$, build a table of values, plot, and join with a smooth curve — never with straight segments.




*The parabola crosses the x-axis at the roots and turns at the vertex.*

WHAT TO READ OFF THE GRAPH

- **Roots** — where the curve meets the $x$-axis.
- **y-intercept** — the value $c$.
- **Axis of symmetry** — the vertical line $x = -\frac{b}{2a}$.
- **Vertex** — on that line; the lowest point if $a > 0$, the highest if $a < 0$.

$a > 0$ gives a curve opening upwards (a “valley”); $a < 0$ opens downwards (a “hill”). The graph is a picture of everything the discriminant told you.

## 02 Worked examples

**EXAMPLE 1** **A population of 50 000 grows 3% a year. Find it after 4 years.**

1. $multiplier = 1.03$

   A 3% increase.

2. $50\ 000 \times 1.03^4$

   Four years.

3. $= 50\ 000 \times 1.1255$

4. $\approx 56\ 275$

ANSWER

$\approx 56\ 300$ people

**EXAMPLE 2** **Solve $3x + 2y = 16$ and $x - y = 3$**

1. $x = y + 3$

   Make x the subject of the second.

2. $3(y + 3) + 2y = 16$

   Substitute.

3. $5y = 7, y = 1.4$

4. $x = 4.4$

   Check: $3(4.4) + 2(1.4) = 16$ ✓

ANSWER

$x = 4.4, y = 1.4$

EXAMPLE 3 ***For $y = x^2 - 2x - 3$, find the roots, the y-intercept and the vertex.***

1. $x^2 - 2x - 3 = 0 \Longrightarrow (x - 3)(x + 1) = 0$

   Roots $3$ and $-1$.

2. $y\text{-intercept: } c = -3$

   Put $x = 0$.

3. $\text{axis: } x = -\dfrac{-2}{2} = 1$

4. $y = 1 - 2 - 3 = -4$

   Vertex $(1; -4)$.

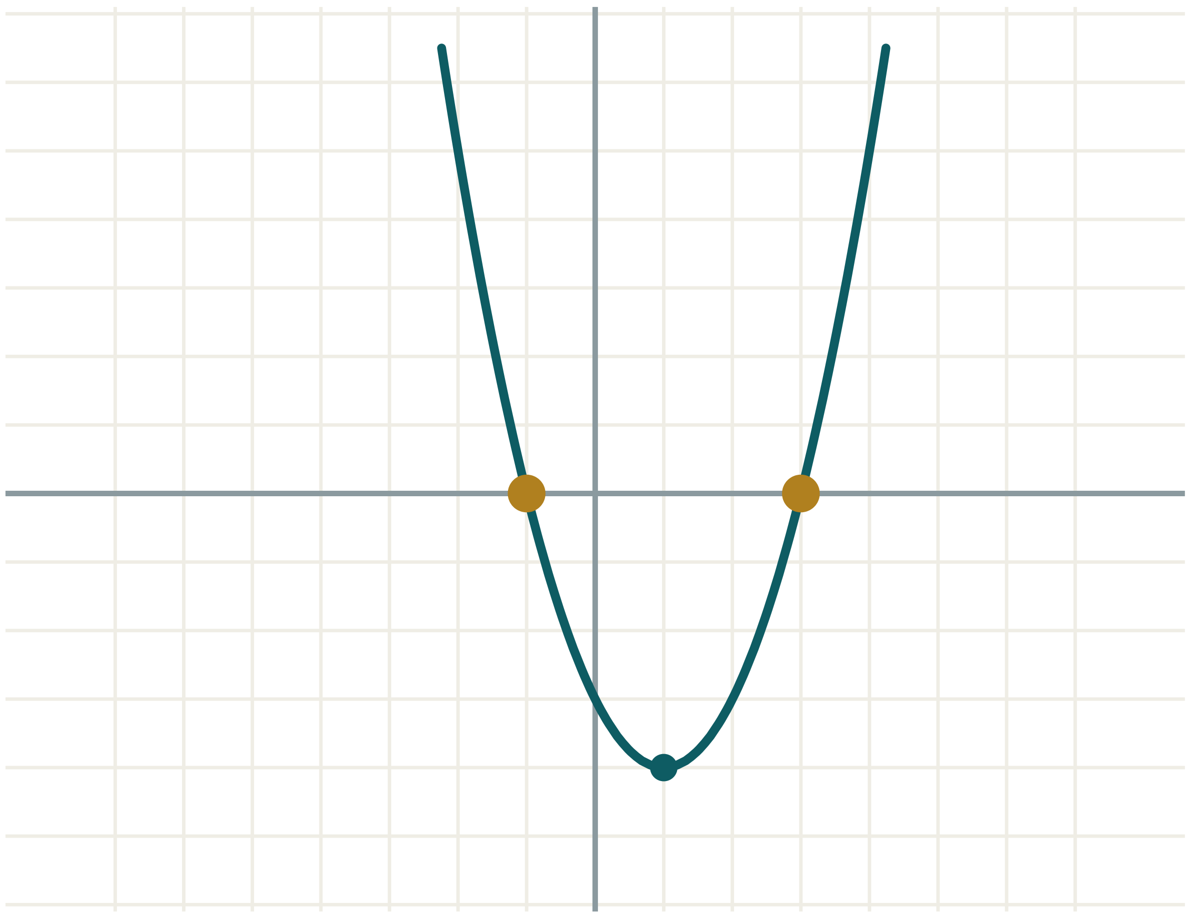
ANSWER

Roots $-1, 3$; intercept $-3$; vertex $(1; -4)$

## 03 Interactive model

Change a, b and c and let the class predict the vertex before the readout shows it.

### Reading a parabola



a **1** | b **-2** | c **-3** | $D = b^2 - 4ac$ **16** | roots **two** | $x_1$ **-1** | $x_2$ **3** | $x_1 + x_2$ **2** | $x_1 \cdot x_2$ **-3**

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| Compound percentage | Murakkab foiz | Сложный процент |
| Multiplier | Ko‘paytiruvchi | Множитель |
| Increase / decrease | Ortish / kamayish | Увеличение / уменьшение |
| Simultaneous equations | Tenglamalar sistemasi | Система уравнений |
| Substitution method | O‘rniga qo‘yish usuli | Метод подстановки |
| Elimination method | Qo‘shish usuli | Метод сложения |
| Table of values | Qiymatlar jadvali | Таблица значений |
| Vertex of a parabola | Parabola uchi | Вершина параболы |
| Axis of symmetry | Simmetriya o‘qi | Ось симметрии |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

# 05 Quick check

### Check yourself

A 10% rise then a 10% fall leaves you with:

- the same
- 1% more
- 1% less
- 10% less

Three years at 5% multiplies by:

- $1.15$
- $1.05^3$
- $3 \times 1.05$
- $0.95^3$

The axis of symmetry of $y = x^2 - 6x + 5$ is:

- $x = 3$
- $x = -3$
- $x = 6$
- $x = 5$

Two simultaneous equations whose lines are parallel have:

- one solution
- no solution
- two solutions
- infinitely many

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up 7 PROBLEMS

1. *Increase 200 by 10%*

   ANSWER *220*

2. *Decrease 200 by 10%*

   ANSWER *180*

3. *Write the multiplier for a 7% increase*

   ANSWER *1.07*

4. *Write the multiplier for a 15% decrease*

   ANSWER *0.85*

5. Solve $x + y = 7$ and $x - y = 1$

   ANSWER $x = 4, y = 3$

6. Solve $y = 2x$ and $x + y = 9$

   ANSWER $x = 3, y = 6$

7. Find the y-intercept of $y = x^2 + 3x - 4$

   ANSWER $-4$

## Medium · the standard question

7 PROBLEMS

1. *A population of 50 000 grows 3% a year. Find it after 4 years.*

   ANSWER $\approx 56\ 300$

2. *A car worth 30 000 000 so‘m loses 12% a year. Find its value after 3 years.*

   ANSWER $\approx 20\ 440\ 000$

3. *A 20% rise is followed by a 20% fall. Find the overall change.*

   ANSWER *4%* decrease

4. Solve $3x + 2y = 16$ and $x - y = 3$

   ANSWER $x = 4.4, y = 1.4$

5. Solve $2x + 3y = 12$ and $2x - y = 4$

   ANSWER $x = 3, y = 2$

6. Find the roots and vertex of $y = x^2 - 2x - 3$

   ANSWER Roots $-1, 3$; vertex $(1; -4)$

7. Find the axis of symmetry of $y = 2x^2 - 8x + 1$

   ANSWER $x = 2$

## Hard · stretch and reason

7 PROBLEMS

1. *A sum grows to 1.331 times its value in 3 years at a constant rate. Find the rate.*

   ANSWER $r^3 = 1.331$, so $r = 1.1$ — 10% a year.

2. *A price falls 25% then rises 25%. Find the overall change.*

   ANSWER $0.75 \times 1.25 = 0.9375$ — a 6.25% loss.

3. *Solve* $4x - 3y = 11$ *and* $3x + 2y = 4$

   ANSWER $x = 2, y = -1$

4. *Two numbers add to 30 and differ by 8. Find them, using simultaneous equations.*

   ANSWER $19$ and $11$

5. *Sketch* $y = -x^2 + 4x$ *and give its roots and vertex.*

   ANSWER Roots $0$ and $4$; vertex $(2; 4)$; opens downwards.

6. *For which values of* $c$ *does* $y = x^2 - 4x + c$ *touch the* $x$*-axis?*

   ANSWER $D = 0$ gives $c = 4$.

7. *A shop raises a price by 10% each year for 2 years, then cuts it by 21%. Show the price returns to the start.*

   ANSWER $1.1^2 \times 0.79 = 1.21 \times 0.79 \approx 0.956$ — close, but a 4.4% loss; an exact return needs a cut of $17.36\%$.

## 07 Homework

### Homework — 6 problems

Algebra 8, pp. 159–164, and Cambridge Learner’s Book 3.3, 4.2, 10.2.

1. *A town of 80 000 grows 2% a year. Find its population after 5 years.*
2. *A machine worth 12 000 000 so‘m loses 15% a year. Find its value after 4 years.*
3. *Solve* $5x + 2y = 21$ *and* $3x - y = 4$
4. *Solve* $2x + y = 10$ *and* $x - 3y = -9$

5. *Draw the graph of* $y = x^2 - 4x + 3$ *for* $x$ *from* $-1$ *to* $5$*, and read off the roots.*
6. *Find the vertex and the axis of symmetry of* $y = x^2 + 6x + 5$

# Control work 6 · Quadratic equations

The Chapter III assessment, and the four errors it produces every year.

LESSON 77–78 | 2 × 40 MIN | ALGEBRA 8, CHAPTER III | STAGE 9 · 4.1

LESSON CLOCK

2' | 36' | 2'

| | |
|---|---|
| Setting up | 2 min |
| The paper | 36 min |
| Collect in | 2 min |

LEARNING OBJECTIVES

- Assess incomplete equations, the discriminant, the formula, Vieta and word problems.
- Diagnose each lost mark by error type.
- Re-solve every task that was lost.

TEXTBOOK

| | |
|---|---|
| National | Revision of Chapter III |
| Cambridge | Learner's Book pp. 84–95 |
| Workbook | Workbook 4.1 |

## 01 Explanation

### Lesson 77 — the paper (40 minutes)

Two variants of seven tasks, 2 marks each:

- task 1 · an incomplete quadratic
- task 2 · solve by factorising
- task 3 · find D and say how many roots
- task 4 · solve with the formula
- task 5 · Vieta — find a missing coefficient or root
- task 6 · a biquadratic equation
- task 7 · a word problem

**Marking note:** task 7 loses a mark if the impossible root is not rejected in writing.

### Lesson 78 — work on mistakes (40 minutes)

THE FOUR ERRORS THIS PAPER PRODUCES

1. **Sign slips in D** — writing $-4ac$ as negative when $c$ is already negative.
2. **Dividing by x**, losing the root $x = 0$.
3. **Negative t kept** in a biquadratic, producing imaginary “roots”.

4. **The impossible root not rejected** in a word problem.

## 02 Worked examples

**EXAMPLE 1** **Find the error: for $x^2 - 2x - 3 = 0$, $D = 4 - 12 = -8$**

1. $c = -3$, so $-4ac = -4 \cdot 1 \cdot (-3) = +12$.
   Two minus signs make a plus.

2. $D = 4 + 12 = 16$

3. $x = \dfrac{2 \pm 4}{2} = 3$ or $-1$

ANSWER

$D = 16$, roots $3$ and $-1$

**EXAMPLE 2** **Find the error: $x^2 = 4x$, so $x = 4$**

1. Both sides were divided by $x$ — which may be zero.

2. $x^2 - 4x = 0 \Longrightarrow x(x - 4) = 0$
   Factorise instead.

3. $x = 0$ or $x = 4$
   The root 0 was lost.

ANSWER

$x = 0, x = 4$

## 03 Interactive model

Show the wrong working, take a vote on the error type, then reveal.

**Diagnose the error**

**Claimed:** for $x^2 - 2x - 3 = 0$, $D = 4 - 12 = -8$

1. $a = 1, b = -2, c = -3$

Read the signs off carefully.

2. $-4ac = -4 \cdot 1 \cdot (-3) = +12$

The two minus signs cancel.

3. $D = 4 + 12 = 16 > 0$

Two roots, not none.

4. $x = 3$

or $x = -1$

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
| --- | --- | --- |
| Control work | Nazorat ishi | Контрольная работа |
| Work on mistakes | Xatolar ustida ishlash | Работа над ошибками |
| Discriminant | Diskriminant | Дискриминант |
| Root | Ildiz | Корень |
| Reject | Rad etish | Отбросить |
| Check by substitution | O‘rniga qo‘yib tekshirish | Проверка подстановкой |
| Word problem | Matnli masala | Текстовая задача |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

# 05 Quick check

### Check yourself

For $x^2 - 2x - 3 = 0$, D is:

- $-8$
- $16$
- $4$
- $-4$

$x^2 = 4x$ has roots:

- $x = 4$
- $x = 0, 4$
- $x = \pm 2$
- $x = 0$

In a biquadratic, $t = -4$ gives:

- $x = \pm 2$
- no real x
- $x = 2$
- $x = -2$

A width comes out as $5$ or $-8$. The answer is:

- both
- $-8$
- $5$, rejecting $-8$
- neither

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up 7 PROBLEMS

1. ***Variant 1, task 1.*** *Solve* $x^2 - 36 = 0$

   ANSWER $x = \pm 6$

2. ***Task 2.*** *Solve* $x^2 - 6x + 8 = 0$

   ANSWER $x = 2, x = 4$

3. ***Task 3.*** *Find D for* $x^2 + 2x + 5 = 0$ *and say how many roots.*

   ANSWER $D = -16$, none

4. ***Task 4.*** *Solve* $2x^2 - 5x + 2 = 0$

   ANSWER $x = 2, x = 0.5$

5. ***Task 5.*** *One root of* $x^2 + bx - 12 = 0$ *is 4. Find b.*

   ANSWER other root $-3$, $b = -1$

6. ***Task 6.*** *Solve* $x^4 - 5x^2 + 4 = 0$

   ANSWER $x = \pm 1, \pm 2$

7. ***Task 7.*** *A rectangle is 2 m longer than wide, area 35 m². Find its sides.*

   ANSWER $5\ m \times 7\ m$

## Medium · the standard question

7 PROBLEMS

1. ***Variant 2, task 1.*** *Solve* $3x^2 - 12x = 0$

   ANSWER $x = 0, x = 4$

2. ***Task 2.*** *Solve* $x^2 + x - 20 = 0$

   ANSWER $x = 4, x = -5$

3. ***Task 3.*** *Find D for* $2x^2 - 4x + 2 = 0$ *and say how many roots.*

   ANSWER $D = 0$, one

4. ***Task 4.*** *Solve* $3x^2 + 5x - 2 = 0$

   ANSWER $x = \frac{1}{3}, x = -2$

5. ***Task 5.*** *The roots of* $x^2 + px + q = 0$ *are 2 and −7. Find p and q.*

   ANSWER $p = 5, q = -14$

6. ***Task 6.*** *Solve* $x^4 - 10x^2 + 9 = 0$

   ANSWER $x = \pm 1, \pm 3$

7. ***Task 7.*** *Two consecutive whole numbers have product 210. Find them.*

   ANSWER $14$ and $15$

## Hard · stretch and reason

7 PROBLEMS

1. *Find the error: for* $x^2 - 2x - 3 = 0$, $D = 4 - 12 = -8$

   ANSWER $-4ac = +12$; correct $D = 16$, roots $3, -1$.

2. *Find the error:* $x^2 = 4x$, *so* $x = 4$

   ANSWER Dividing by $x$ lost the root $0$.

3. *Find the error:* $x^4 + 3x^2 - 4 = 0$ *gives* $x = \pm 1, \pm 2$

   ANSWER $t = -4$ must be rejected; only $x = \pm 1$.

4. *Find the error: a width comes out 5 or −8, answer "5 and −8"*

   ANSWER A length cannot be negative — the rejection must be stated.

5. *Find the error: "*$x^2 = 16$, *so* $x = 4$*"*

   ANSWER The negative root was lost: $x = \pm 4$.

6. *Find the error: "*$(x - 1)(x - 4) = 6$, *so* $x - 1 = 6$ *or* $x - 4 = 6$*"*

   ANSWER The zero product rule needs a zero on the right; expand and re-solve.

7. *Find the error: "*$D = 25$, *so* $x = \dfrac{-b \pm 25}{2a}$*"*

   ANSWER The formula uses $\sqrt{D} = 5$, not $D$ itself.

## 07 Homework

### After the work-on-mistakes lesson

Re-solve every task you lost marks on. Quarter IV opens with statistics.

1. *Solve* $2x^2 - 7x - 4 = 0$
2. *Solve* $x^4 - 13x^2 + 36 = 0$
3. *One root of* $x^2 + bx + 15 = 0$ *is 5. Find b and the other root.*
4. *A rectangle is 4 cm longer than wide and has area 96 cm². Find its sides.*

5. Write out the four errors from this lesson with one example each.